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Comparisons of the Effectiveness and Safety of Surgical Treatment for Neovascular Glaucoma: A Systematic Review and Network Meta-Analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-051794
Article Type:	Original research
Date Submitted by the Author:	31-Mar-2021
Complete List of Authors:	Lin, Peijie; Jinan University, The 2rd College of Clinical Medicine Zhao, Qian; Shenzhen Eye Hospital He, Jing; Shenzhen Eye Hospital Fan, Wei; Jinan University, The 2rd College of Clinical Medicine He, Wenyi; Jinan University, The 2rd College of Clinical Medicine Lai, Mingying; Shenzhen Eye Hospital
Keywords:	Ophthalmology < SURGERY, Medical ophthalmology < OPHTHALMOLOGY, Glaucoma < OPHTHALMOLOGY

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Comparisons of the Effectiveness and Safety of Surgical Treatment for Neovascular Glaucoma: A Systematic Review and Network Meta-Analysis

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Running title: Network Meta-Analysis of Surgery for Neovascular Glaucoma

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Sources of support:

Supported by Science and Technology Plan Projects of Shenzhen, China

[JCYJ20160428144848002].

Word count: 3420 words.

Number of figures and tables: 5 figures and 3 tables.

Conflicts of interest:

All authors have completed the ICMJE uniform disclosure form. The authors have no conflicts of interest to declare.

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Abstract

Objectives: To compare the effectiveness and safety of the six interventions for neovascular glaucoma.

Design: A systematic review and network meta-analysis.

Setting: None.

Participants: None.

Methods: Randomized controlled trials and cohort studies which compared the six interventions in neovascular glaucoma were identified using the following databases: PubMed, Cochrane Library, Embase, Web of Science. Network meta-analysis was performed using STATA version 15.0.

Primary and secondary outcome measures: The primary one was the weighted mean differences (WMDs) for intraocular pressure reduction. Secondary one was odds ratios (ORs) for success rate. Outcome measures were reported with a 95% confidence interval (CI) and P< 0.05 was considered statistically significant.

Results: Twenty-three studies involving a total of 1303 patients were included. Network meta-analysis showed that in comparison with AGV, AGV+IVAV(MD=4.74, 95%[1.04, 8.45]) and Trab(MMC)+IVAV(MD=6.19, 95%[0.99, 11.40]) showed a favorable effect in IOPR 6 months after surgery. Compared with CCT, AGV (MD=1.80, 95%[0.63, 2.97]), AGV+IVAV (MD=2.33, 95%[1.19, 3.48]), CPC (MD=2.09, 95%[0.63, 3.55]), Trab(MMC) (MD=1.26, 95%[0.14, 2.39]),

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Trab(MMC)+IVAV (MD=1.75, 95%[0.83, 2.68]) showed a superior impact in success rate. The order of efficacy as best intervention ranked as follows: Trab(MMC)+IVAV (IOPR 6 months after surgery, SUCRA=88.1), CPC (IOPR 12 months after surgery, SUCRA=81.9), AGV+IVAV (IOPR 12 months after surgery, SUCRA=79.9), AGV+IVAV (success rate, SUCRA=92.7).

Conclusion: In the treatment of neovascular glaucoma, AGV+IVAV and CPC were superior to the other four interventions**.** Additionally, AGV+IVAV is superior to CPC concerning the success rate in the long-term treatment. However, considering the limitations of this review, more high-quality trials, especially those surgical interventions not mentioned in this review, should be carried out in the future to further confirm the current findings.

Abbreviations: IOPR, intraocular pressure reduction; AGV, ahmed glaucoma valve implant surgery; CPC, cyclophotocoagulation; CCT, cyclocryotherapy; Trab(MMC), trabeculectomy with mitomycin; IVAV, intravitreal anti-vascular endothelial growth factor.

Trial registration: None.

Keywords: Ophthalmology; Medical ophthalmology; Glaucoma

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Article Sumary: Strengths and limitations of this study

- ✓ To the best of our knowledge, this is the most comprehensive network meta-analysis which includes all the available data of comparative studies and evaluates different aspects of different therapeutic strategies.
- ✓ The effectiveness and the safety were assessed with rigorous inclusion criteria, leading to more convincing results.
- ✓ In terms of intraocular pressure reduction, we selected the common follow-up time point between all studies to reduce bias, including 6 months and 12 months.
- ✓ There was no common time point of data concerning success rate, we used information available at the final follow-up for statistical analysis.

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1 Introduction

Neovascular glaucoma (NVG) is a potentially blinding, refractory glaucoma,[1] which is characterized by neovascularization (NV) in iris or anterior chamber angle.[2] NVG manifests as dramatic elevation of intraocular pressure (IOP), severe ocular pain and vision loss, influencing the quality of life in patients adversely. It is secondary to various ocular ischemic diseases, such as diabetic retinopathy, central retinal vein occlusion, and ocular ischemic syndrome.[2] With a high prevalence of diabetes and vascular disease, the incidence of NVG is increasing steadily, accounting for more than 30% of refractory glaucoma and becomes an occupational health issue.[1] Medical intervention alone usually did not show noteworthy benefit, surgery is regarded as the first-line choice for NVG, but complicated by higher failure rates and more difficult tissue anatomy than in primary glaucoma.[3] Thus, its treatments have been paid high attention to in glaucoma.

Currently, glaucoma drainage devices (GDDs),[4] filtering surgery,[5] cyclodestructive surgery,[6] panretinal photocoagulation (PRP) or intravitreal vascular endothelial growth factor (VEGF) inhibitors,[7] are routinely utilized surgical treatment modalities for NVG,[8] which present with a certain effectiveness. Many authors verified the effectiveness of these treatment modalities.[9, 10] Nevertheless, there is no sufficient evidence of superiority of one over another. Meanwhile, the combined effects of surgery and anti-VEGF drugs also play an important role in NVG treatment, which further increases the diversity of treatment strategies.[11] Some systematic review and

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meta-analyses have been published to evaluate the effectiveness and safety of these combination of treatment modalities for NVG. Two meta-analyses reported the effect of intravitreal bevacizumab injection before AGV implantation, which indicated higher surgical success rate.[12, 13] However, because of the limitations of the traditional pair-wise meta-analysis, it is difficult to determine which one is the best management of NVG in the modalities.

In recent years, network meta-analysis has been developed. Compared with traditional pairwise meta-analysis, it allows for combining of data related to multiple treatments, comparison of interventions based on indirect information, and generating a ranking of treatment arms according to the efficacy.[14] A latest network meta-analysissuggested that trabeculectomy with mitomycin and interferon was the most likely to improve the success rate in treatment of NVG,[15] followed by glaucoma valve; however, there was an inevitable insufficiency of evidence to determine the optimal surgical interventions since intraocular pressure reduction and adverse events was not mentioned in the analysis. In addition, the result of Loop-specific approach showed a certain local inconsistency signifying that the findings related to trabeculectomy should be interpreted cautiously.

Therefore, aiming to add more sufficient evidence, we performed a network meta-analysis involving more outcome measurements to evaluate the comparative effectiveness and safety of different surgical treatment, to help ophthalmologist better

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make treatment strategies for patients with NVG. We present the following article in accordance with the PRISMA reporting checklist.

2 Material and methods

2.1 Data Sources

A medical literature search was performed in the following databases from their inception through September 1st, 2020: PubMed, Web of Science, the Cochrane Library, Embase. Different from Dong's network meta-analysis,[15] we neglected the Chinese database due to the lower quality of literature retrieved. There are no restrictions regarding publication year, language, or methodological filter on these studies. The searching was performed using medical subject headings (MeSH) and keywords, including "neovascular glaucoma", "neovascular", "rubeosis and iris", "rubeosis iridis", "neovascularization and iris", "rubeotic glaucoma", "congestive and glaucoma", "haemorrhagic", "hemorrhagic", "NVG", "NVI" and "NVA". The details of search strategy are showed in Appendix (page 2) (available here). Two reviewers searched literatures, reviewed the titles and abstracts of articles independently to select the potential ones. The full texts of the selected articles were checked based on inclusion and exclusion criteria, and the divergent articles were checked by a third reviewer.

2.2 Inclusion Criteria

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Studies were included according to the following criteria: (1) study type: randomized controlled trials (RCTs) or cohort studies; (2) population: patients were diagnosed as NVG by symptoms, signs and examination; (3) intervention: controlled study of different therapeutic strategies related to different surgeries; (4) outcome variables: at least one of the outcome measures was required.

2.3 Exclusion Criteria

The following studies were excluded if (1) follow-up time was < 6 months; (2) trials with a small sample size of (n < 10); (3) drug dose-related study; (4) comparative studies of similar surgical procedures, such as trabeculectomy versus modified trabeculectomy; (5) abstracts from conferences and full texts without raw data available for retrieval, literature reviews, letters and duplicate publications.

2.4 Outcome Measure

The primary outcome was the mean difference in intraocular pressure reduction (IOPR) after surgery, with or without anti-glaucoma medication. When authors reported the mean and SD of the IOPR, we used these directly. If not available, we computed them as follows: $IOPR = IOP_{baseline} - IOP_{endpoint}$, $SD_{IOPR} = (SD^2_{baseline} + SD^2_{endpoint} - SD^2_{baseline} \times SD^2_{endpoint})^{1/2}$. In order to reduce bias, we preferably selected the common follow-up time point between all studies: mean IOPR 6 months after surgery and mean IOPR 12 months after surgery. In case 12-month datas were not reported, we used the latest reported outcome data or the data closest to 12 months.[16] The secondary outcomes

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were the rates of surgical success, using the definitions used by authors of individual studies. In case there was no common time point of data concerning success rate, we used information available at the final follow-up for statistical analysis.

2.5 Data Extraction

Two investigators independently worked for data extraction, and they collected the following information: (1) basic characteristics, including author name, study design, publication year, age and gender of patients, intervention, sample size, stages of NVG, outcomes, and follow-up; (2) clinical outcomes, including IOPR 6 months after surgery, IOPR 12 months after surgery, and success rate.

2.6 Quality assessment

The quality assessment of RCTs was conducted by using the Cochrane risk of bias tool, and cohort studies using the Newcastle-Ottawa scale. Quality assessment was independently carried out by two reviewers, and disagreements were checked by a third reviewer.

2.7 Statistical analysis

A network meta-analysis was carried out using STATA version 15.0. Continuous variables (IOPR 6 months after surgery and IOPR 12 months after surgery) were analyzed using mean difference (MD) and its 95% credible interval (CI), while dichotomous variables (success rate) using Odds Ratio (OR). At the beginning of our network meta-analysis, pair-wise meta-analyses were performed, then “mvmeta”

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package was used to perform the plots of different comparisons, the rankplots based on probabilities and the surface under cumulative ranking (SUCRA) for different endpoints. Furthermore, node-splitting analysis and loop-specific approach were used to evaluate inconsistency, and the Grades of Recommendations Assessment, Development and Evaluation (GRADE) was used to evaluate the importance of the outcomes.

3 Results

3.1 Identification of the relevant studies

9658 articles were identified in our initial search, from which 2624 articles were excluded for duplications, and 6606 were excluded by reading titles and abstracts. In the remaining 428 articles, full texts were obtained to check eligibility, in which 39 studied were excluded because of non-comparative study, 168 studies were excluded for wrong intervention or comparator, and 198 studies were excluded because of lacking of clinical data. Finally, 23 studies were included in our final analysis. Figure 1 shows the selection process for relevant studies.

3.2 Characteristics of the included studies

The review involved 1303 patients and the sample size ranged from 20 to 170 cases. 4 studies were RCTs and 19 were cohort studies. 22 studies were two-arm studies, and 1 was three-arm study. In our study, the types of surgical treatments included AGV, AGV+IVAV, CPC, CCT, Trab(MMC) and Trab(MMC)+IVAV. 8 studies compared AGV with AGV+IVAV, 3 studies compared AGV with CPC, 3 compared AGV+IVAV

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with Trab(MMC)+IVAV, 2 compared CCT with Trab(MMC)+IVAV, 5 compared Trab(MMC) with Trab(MMC)+IVAV, 1 compared AGV with Trab(MMC) and 1 compared AGV with Trab(MMC)+IVAV. 308 patients were included in AGV groups, 309 patients in AGV+IVAV groups, 67 patients in CPC groups, 53 patients in CCT groups, 235 patients in Trab(MMC) groups, and 355 patients in Trab(MMC)+IVAV groups. In terms of the criteria of IOPR, 14 trials employed the criteria of IOPR 6 months after surgery, 12 trials used the criteria of IOPR 12 months after surgery. As summarized in Appendix (Page 3-6), 21 trials used heterogeneous criteria of success rate defined by authors of individual studies. The baseline characteristics of each study are presented in Table 1.

Table 1. The characteristics of the included studies

References	Study design	Interventions	Age(MD,SD)	Gender (male/female)	Sample size		Outcomes	Mean follow-up(months)	anti-VEGF	Combine PRP
					No. Patients	No. Eyes				
Ma,2012[17]	OCT	AGV vs AGV+IVAV	58.78(13.09) vs 57.60(13.56)	27/25	N/A	32 vs 20	(c)(d)	15	Bevacizumab	Yes
Mahdy,2013[18]	RCT	AGV vs AGV+IVAV	56(4.3) vs 55(1.3)	23/17	20 vs 20	20 vs 20	(a)(b)(c)	18	Bevacizumab	Yes
Zhou,2013[19]	OCT	AGV vs AGV+IVAV	57.89(11.51) vs 54.40(13.68)	36/17	28 vs 25	28 vs 25	(a)(b)(c)	15.36(4.53) vs 15.08(5.14)	Bevacizumab	Yes
Kang,2014[20]	OCT	AGV vs AGV+IVAV	54.3(10.8) vs 54.8(13.0)	22/5	13 vs 14	13 vs 14	(c)	6	Bevacizumab	Yes
Arcieri,2015[21]	RCT	AGV vs AGV+IVAV	62.40(11.78) vs 59.25(8.05)	24/16	20 vs 20	20 vs 20	(a)(b)(c)	28.2(8.04) vs 25.8(8.04)	Bevacizumab	Yes
Kwon,2017[22]	OCT	AGV vs AGV+IVAV	57(18) vs 59(10)	56/14	25 vs 45	25 vs 45	(b)(c)	27(14) vs 26(16)	Bevacizumab	Yes
Li,2019[23]	OCT	AGV vs AGV+IVAV	57.49(8.42) vs 58.06(7.33)	46/34	34 vs 46	34 vs 46	(c)	6	Ranibizumab	N/A

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Kong,2017[24]	OCT	AGV vs AGV+IVAV vs AGV+IVAV	54.24(11.05) vs 53.00(12.32) vs 54.00(12.74)	43/25	21 vs 26 vs 21	21 vs 26 vs 21	(a)(b)(c)	12	Ranibizumab; Conbercept	N/A
Yildirim,2009[25]	RCT	AGV vs CPC	57.2(10.3) vs 60.0(11.7)	35/31	33 vs 33	33 vs 33	(a)(b)(c)	24	No	N/A
Liu,2014[26]	OCT	AGV vs CPC	57.30(12.88) vs 56.44(13.16)	27/19	24 vs 22	31 vs 26	(a)	6	No	N/A
Choy,2018[27]	RCT	AGV vs CPC	62.8(11.0) vs 61.3(13.5)	13/6	12 vs 8	12 vs 8	(b)(c)	31.0(15.4) vs 28.5(17.9)	No	Yes
Zhang,2014[28]	OCT	AGV+IVB vs Trab(MMC)+IVAV	57.5(12.5) vs 57.0(11.5)	12/10	12 vs 10	13 vs 10	(b)	26	Bevacizumab	Yes
Sun,2017[29]	OCT	AGV+IVR vs Trab(MMC)+IVAV	52.42(12.78)	23/22	23 vs 22	23 vs 22	(a)(b)(c)	12	Ranibizumab	Yes
Gao,2018[30]	OCT	AGV+IVR vs Trab(MMC)+IVAV	56.25(9.98) vs 54.75(11.10)	49/27	36 vs 40	36 vs 40	(a)(b)(c)	12	Ranibizumab	Yes
Du,2015[31]	OCT	CCT vs Trab(MMC)+IVAV	57. 6(13. 4)	31/15	23 vs 23	23 vs 23	(a)(b)(c)	12	Bevacizumab	Yes
Wang,2016[32]	OCT	CCT vs Trab(MMC)+IVAV	45.5(4.6) vs 44.5(6.3)	33/27	30 vs 30	30 vs 30	(a)(b)(c)	12(0.3)	Ranibizumab	Yes
Saito,2010[33]	OCT	Trab(MMC) vs Trab(MMC)+IVAV	61(14) vs 60(13)	33/19	32 vs 20	32 vs 20	(c)	28(13) vs 12(5)	Bevacizumab	Yes
Takihara,2011[34]	OCT	Trab(MMC) vs Trab(MMC)+IVAV	60.0(10.4) vs 60.3(11.6)	44/6	30 vs 20	33 vs 24	(a)(b)(c)	25.3(9.8) vs 15.2(3.6)	Bevacizumab	Yes
Lee,2015[35]	OCT	Trab(MMC) vs Trab(MMC)+IVAV	52.73(11.27) vs 56.53(11.86)	38/20	26 vs 32	26 vs 32	(c)	12	Bevacizumab	Yes
Yu,2018[36]	OCT	Trab(MMC) vs Trab(MMC)+IVAV	54.88(4.19) vs 55.71(4.23)	48/42	42 vs 48	42 vs 48	(a)(c)	6	Conbercept	Yes
Song,2019[37]	OCT	Trab(MMC) vs Trab(MMC)+IVAV	59.25(5.51) vs 58.73(5.44)	91/79	82 vs 88	82 vs 88	(a)	6	Conbercept	N/A
Shen,2011[38]	OCT	AGV vs Trab(MMC)	54.0(15.6) vs 59.65(15.8)	19/21	20 vs 20	20 vs 20	(b)(c)	31.1(24.5) vs 25.0(19.7)	No	N/A
Liu,2016[39]	OCT	AGV vs Trab(MMC)+IVAV	56.7(13.6) vs 62.3(10.8)	21/16	19 vs 18	19 vs 18	(a)(c)	6	Ranibizumab	N/A

Notes: OCT, observational cohort study; RCT, randomized controlled trial; PRP, panretinal photocoagulation; AGV, ahmed glaucoma valve implant surgery; CPC, cyclophotocoagulation; CCT, cyclocryotherapy; MMC, mitomycin; Trab(MMC), trabeculectomy with MMC; VEGF, vascular endothelial growth factor; IVAV, intravitreal anti-VEGF; AGV+IVAV, AGV combined with IVAV; Trab(MMC)+IVAV, Trab(MMC) combined with IVAV;(a), intraocular pressure reduction 6 months after surgery;(b), intraocular pressure reduction 12 months after surgery;(c), success rate

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3.3 Quality Assessment

In 4 RCTs, one studies didn't state the method of randomization in details, three study employed computer of random number for randomization. Only one study mentioned the blinding method. All the RCTs had complete data. The risk of bias summary is shown in Appendix (Page 7). In terms of the cohort studies, based on the New Castle-Ottawa quality assessment scale, 1 studies scored 9 points, 8 studies scored 8 points, 5 studies scored 7 points and 5 studies scored 6 points (Appendix).

3.4 The Results of Meta-Analysis

The results of pair-wise meta-analysis are demonstrated in Appendix (Page 10). Figure 2 shows the network of eligible comparisons for IOPR 6 months after surgery, IOPR 12 months after surgery and success rate, and Figure 3 shows the results of network meta-analysis.

Fourteen studies involving 882 eyes reported IOPR 6 months after surgery (Figure 2). When compared with AGV, AGV+IVAV and Trab(MMC)+IVAV showed a significantly higher IOPR 6 months after surgery, but no significant difference was found between AGV+IVAV and Trab(MMC)+IVAV. Meanwhile, no significant difference was found among the other interventions.

13 studies involving 681 eyes and 14 comparisons reported IOPR 12 months after surgery (Figure 2). As illustrated in Figure 3, no significant difference was found in IOPR 12 months after surgery.

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In terms of success rate, 20 studies involving 21 comparisons and 1098 eyes were merged for analysis (Figure 2). Compared with CCT, the interventions including AGV, AGV+IVAV, CPC, Trab(MMC) and Trab(MMC)+IVAV presented with a significantly higher success rate. Additionally, the success rate in AGV+IVAV was significantly higher than that in AGV and Trab(MMC), and no significant difference was found between the latter two treatments. However, no significant difference was found among the other interventions.

The plots of probability and SUCRA are illustrated in Appendix. Table 2 shows that Trab(MMC)+IVAV had the highest probability to be the best intervention in IOPR 6 months after surgery while AGV+IVAV ranking behind. CPC and AGV+IVAV had the highest probability to be the best intervention in IOPR 12 months after surgery, and AGV+IVAV had the highest probability to be the best intervention in success rate, respectively.

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Table 2. The results of SUCRA and probability.

Treatments/Outcomes	SUCRA	PrBest	MeanRank
IOPR (6mo)			
AGV	14.5	0.1	5.3
AGV+IVAV	68.9	21.3	2.6
CPC	30.1	5.4	4.5
CCT	39.6	5.7	4
Trab(MMC)	58.8	12.8	3.1
Trab(MMC)+IVAV	88.1	54.7	1.6
IOPR (12mo)			
AGV	48.2	1.4	3.6
AGV+IVAV	79.9	30.4	2
CPC	81.9	59	1.9
CCT	18.2	2.2	5.1
Trab(MMC)	28.2	4.2	4.6
Trab(MMC)+IVAV	43.6	2.9	3.8
Success rate			
AGV	54.3	0.2	3.3
AGV+IVAV	92.7	66.3	1.4
CPC	73.7	30.3	2.3
CCT	0.4	0	6
Trab(MMC)	25	0.2	4.8
Trab(MMC)+IVAV	53.9	3	3.3

Notes: data are probability in the rows of "SUCRA" and "PrBest". SUCRA= surface under the cumulative ranking, PrBest= the best probability, IOPR (6mo)= intraocular pressure reduction 6 months after surgery, IOPR (12mo)= intraocular pressure reduction 12 months after surgery.

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3.5 Consistency Analysis

Node-splitting analysis was performed to evaluate the inconsistency by comparing direct and indirect effects, indicating no significant inconsistency (Appendix) and the results were reliable. In addition, the results of Loop-specific approach showed no significant inconsistency in the comparisons of closed circles in outcomes of IOPR(12mo) and success rate, but significant inconsistency in IOPR(6mo) (Table 3).

Table 3. Loop-specific approach.

Outcome	Loop	ROR	Z_value	P_value	95%CI	Loop_Heterog_tau2
IOPR(6mo)	AGV-AGVI-TrabI	6.490	0.85	0.395	(0.00,21.45)	25.666
IOPR(12mo)	AGV-AGVI-Trab-TrabI	0.123	0.013	0.990	(0.00,19.05)	24.326
Success rate	AGV-Trab-TrabI	1.574	1.343	0.179	(0.00,3.87)	0
	AGV-AGVI-TrabI	1.282	1.16	0.246	(0.00,3.45)	0.066

Notes: loop-specific approach is used to check the inconsistency which aims at the closed loop. In this analysis, ROR is close to 1, indicating no significant difference between direct and indirect effects. AGVI= AGV+IVAV , Trab= Trab(MMC), TrabI= Trab(MMC)+IVAV.

3.6 GRADE for the Outcome Measurements

We summarized the GRADE judgements in Appendix. According to the suggestions of GRADE workgroups, we combined the evidences of direct and indirect comparisons and chose a higher level, and the results demonstrated the evidences provided in this review were moderate, low or very low.

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3.7 Adverse Events

In the included studies, 22 reported adverse events. Twenty studies reported 225 cases of hyphema, in which 70 occurred in the AGV group, 67 in the Trab(MMC) group, 38 in the AGV+IVAV group, 29 in the Trab(MMC)+IVAV group, 13 in the CCT group and 8 in the CPC group. Ten studies reported 73 cases of corneal oedema, in which 31 occurred in the CCT group, 20 in the AGV group, 14 in the CPC group, 5 in the AGV+IVAV group and 3 in the Trab(MMC) group. Eleven studies reported hypotony, in which 24 eyes occurred in the AGV+IVAV group, 17 in the Trab(MMC)+IVAV group, 16 in the AGV group, 7 in the Trab(MMC) group and 3 in the CPC group. Shallow anterior chamber was reported in 13 studies, including 42 cases in the AGV+IVAV group, 20 in the AGV group, 11 in the Trab(MMC)+IVAV group and 10 in the Trab(MMC) group.

Meanwhile, tube exposure or occlusion was reported in 8 studies, including 23 cases in the AGV group and 17 in the AGV+IVAV group. Encapsulated plate was reported in six studies, including 19 eyes in the AGV group and 12 in the AGV+IVAV group. Bleb leak was reported in six studies, including 6 cases in the Trab(MMC) group, 6 in the Trab(MMC)+IVAV group and 2 in the AGV+IVAV group. Phthisis bulbi was reported in four studies, including 6 eyes in the AGV group and 2 in the Trab(MMC) group.

Moreover, 5 studies reported 24 cases of vitreous hemorrage, in which 10 occurred in the Trab(MMC) group, 7 in the Trab(MMC)+IVAV group, 6 in the AGV group and 1 in the AGV+IVAV group. Choroidal effusion or detachment was reported in 11 studies,

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including 21 cases in the Trab(MMC)+ group, 18 in the Trab(MMC) group, 12 in the AGV group and 10 in the AGV+IVAV group. Two studies reported 2 cases with retinal detachment in the AGV group.

Besides, 9 studies reported anterior segment inflammation, 20 cases occurred in the AGV+IVAV group, 19 in the AGV group, 16 in the CCT group, 11 in the CPC group and 1 in the Trab(MMC)+IVAV group. Ocular pain was reported in five studies, in which 20 cases in the CCT group, 18 in the CPC group, 7 in the AGV group, 5 in the Trab(MMC) group, 2 in the AGV+IVAV group and 2 in the Trab(MMC)+IVAV group. Two studies reported ocular atrophy, 2 cases occurred in the CPC group and 1 in the Trab(MMC) group.

4 Discussion

This is the first network meta-analysis to evaluate the efficacy and safety of the six widely utilized treatment modalities for NVG. Due to similar principles of reducing VEGF, bevacizumab, ranibizumab and conbercept were classified as anti-VEGF inhibitors in our analysis.[40] Since application of panretinal photocoagulation(PRP), laser peripheral iridotomy (LPI), and anti-scarring drugs were common as adjunctive treatments, they could not be compared as stand-alone interventions. Meanwhile, studies with follow-up time which was not mentioned or less than 6 months were excluded. By reason of the foregoing, compared with Dong's network meta-analysis,[15] our inclusion criteria were more rigorous, leading to more convincing results.

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The results demonstrated that in the six interventions, the effectiveness of CCT was the worst, and the effectiveness of AGV+IVAV was better than the other five interventions in success rat, but similar as CPC in IOPR 12 months after surgery. Besides, the effectiveness of Trab(MMC)+IVAV was the best among the six interventions in IOPR 6 months after surgery.

In terms of the rank of probability for the six interventions, it is reasonable that AGV+IVAV demonstrated a highest probability to be the best intervention in success rate. The insertion of GDD implants has been considered as an option with lower risk of failure than conventional filtering surgery for treating NVG.[4] Among GDDs, Ahmed possesses even higher acceptance and popularity and has been widely used by an increasing number of surgeons.[41] In this review, only AGV was included, we neglected other glaucoma valve-related studies in reason that there is no sufficient studies of comparison of each different types of GDDs. In regard to this, several authors conducted reviews and found no evidence of superiority of one over another in treatment of glaucoma.[4, 42, 43] Meanwhile, CPC and AGV+IVAV had an advantage over other four interventions in IOPR 12 months after surgery. The fact that two studies in the CPC groups were RCTs and paitients with similar visual acuity were selected in the non-RCT, avoids create a potential selection bias, since cyclodestructive procedures are normally an option for advanced NVG with limited visual acuity, wheras AGVs are usually implanted in paitents with better prognosis and visual potential.[44] Therefore, AGV+IVAV was superior to CPC concerning the success rate.

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But it is noteworthy that Trab(MMC)+IVAV unexpectedly showed superiority in IOPR 6 months after surgery more than other five interventions. We think the reasons may be attributed to the following aspects. First of all, in this review most of trials have a small sample size, and the number of comparisons was small. As shown in Figure 4, the contribution plot of IOPR 6 months after surgery showed Trab(MMC)+IVAV accounted for a large proportion, demonstrating the small sample size influenced the total effect and final outcomes adversely (Figure 4). In addition, the funnel plots of IOPR 6 months after surgery showed the potential report bias (Figure 5). Moreover, the results of Loop-specific approach showed a significant inconsistency existed in IOPR 6 months after surgery, which means the results of indirect comparisons were not consistent with those of direct ones; two comparisons concluded different conclusions in IOPR 6 months after surgery. Subsequently, the findings in favor of Trab(MMC)+IVAV should be interpreted cautiously. Besides, in respect of IOPR 12 months after surgery and success rate, Trab(MMC)+IVAV showed inferiority to AGV+IVAV , CPC and AGV. As mainstream consideration, it shared a low long-term effectiveness mainly due to neovascular membrane obstruction in the filtering passage or external scarring and conjunctival fibrosis promoted by VEGF, even with antimetabolites such as mitomycin C or 5-fluorouracil.[5, 45]

Our review has two methodological strengths. In this research network meta-analysis was carried out to compare the direct and indirect effect of the four treatments, and the SUCRA plot was performed to estimate the ranks of interventions, which may facilitate

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ophthalmologist to make treatment strategies correctly. However, our review has its disadvantages. First, the evidence from GRADE for included outcomes was relatively low. Only four randomized studies were retrieved, whereas the remaining were non-randomized comparative studies. Second, there was a certain heterogeneity in follow-up time when observing 12-month postoperative IOP. We followed the recommendations of the European Glaucoma Society (European Glaucoma Society, 2017), which states that the latest reported outcome data or the data closest to 12 months could be reported in case 12-month data were not reported.[16] Third, heterogeneous definitions of the success criteria challenges the validity of our analysis and conclusions. Fourth, the number and sample size of retrieved studies are small, which may affect the interpretation of the results. Fifth, only three outcomes were analyzed in our research, more outcomes such as best corrected visual acuity (BCVA) and numbers of anti-glaucoma medications were also relevant but not analyzed, because no sufficient data was reported in the included studies. Finally, the included studies only consisted of six interventions, more surgical treatment should be carried out in the upcoming studies to allow future systematic reviews and meta-analysis. These limitations may affect the final outcomes. In addition, some trials did not mention the adverse events, so the safety of the six treatments could not be evaluated by SUCRA in this review.

5 Conclusion

In conclusion, our review suggested, among the six interventions, AGV+IVAV and CPC were superior to the other four interventions, and the effectiveness of CCT was the

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worst in the treatment of NVG. Additionally, AGV+IVAV is superior to CPC concerning the success rate in the long-term treatment. However, considering the limitations of this study, more high-quality trials, especially those surgical interventions not mentioned in this review, should be carried out in the future to further confirm the current conclusions.

Supplementary Materials

The supplementary appendix has 17 pages, including search strategy, references list of included studies, success and failure criteria applied in each study, risk of bias summary of RCT, the quality assessment according to the NOQAS of each OCT, node-splitting analysis, pair-wise meta-analysis, the surface under the cumulative ranking plots based on cumulative probabilities of interventions, GRADE for outcome measurements. More data can be found in the article. The interested reader can read a supplementary appendix online.

Data availability statement

Data are available in a public, open access repository. All data relevant to the study are included in the article or uploaded as online supplementary information. All included articles could be downloaded from the website https:// pubmed.ncbi.nlm.nih.gov/.

Acknowledgments

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We thank Haoyu Chen from Department of Ophthalmology, Joint Shantou International Eye Center Shantou University and the Chinese University of Hong Kong, for his editorial assistance.

Footnotes

Author contributions: PL conceived the study, drafted the protocol, collected data, performed the statistical analysis and wrote the manuscript. QZ participated in data extraction and data analysis. JH, WF and WH contributed to assembly of data, the quality assessment and data interpretation. ML reviewed and revised the manuscript. All authors have read and approved the final version of the manuscript.

Funding: This work was supported by Science and Technology Plan Projects of Shenzhen, China (grant number: JCYJ20160428144848002).

Competing interests: All authors have completed the ICMJE uniform disclosure form. The authors have no conflicts of interest to declare.

Financial disclose: The authors declare that they have no financial disclose.

Patient and Public Involvement: No patient involved.

Ethical approval and informed consent: Not applicable as no new patients were involved in this research.

Patient consent for publication: Not required.

Provenance and peer review: Not commissioned; externally peer reviewed.

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References

1 Havens SJ, Gulati V. Neovascular Glaucoma. *Dev Ophthalmol* 2016;55:196-204.
2 Shazly TA, Latina MA. Neovascular glaucoma: etiology, diagnosis and prognosis. *Semin Ophthalmol* 2009;24(2):113-21.
3 Hayreh SS. Neovascular glaucoma. *Prog Retin Eye Res* 2007;26(5):470-85.
4 Tseng VL, Coleman AL, Chang MY, Caprioli J. Aqueous shunts for glaucoma. *Cochrane Database Syst Rev* 2017;7:CD004918.
5 Dahan E, Ben Simon GJ. An augmented trabeculectomy for neovascular glaucoma. *Ophthalmic Surg Lasers Imaging* 2011;42(3):196-201.
6 Freigassner P, Eckhardt M. Transscleral cyclophotocoagulation versus cyclocryotherapy in treatment of neovascular glaucoma: a retrospective analysis. *Acta Ophthalmol Scand* 2003;81(6):674-5.
7 Yang H, Yu X, Sun X. Neovascular glaucoma: Handling in the future. *Taiwan J Ophthalmol* 2018;8(2):60-6.
8 Olmos LC, Lee RK. Medical and surgical treatment of neovascular glaucoma. *Int Ophthalmol Clin* 2011;51(3):27-36.
9 Shchomak Z, Cordeiro Sousa D, Leal I, Abegão Pinto L. Surgical treatment of neovascular glaucoma: a systematic review and meta-analysis. *Graefes Arch Clin Exp Ophthalmol* 2019;257(6):1079-89.
10 Lee NY, Hwang HB, Oh SH, Park CK. Efficacy of Additional Glaucoma Drainage Device Insertion in Refractory Glaucoma: Case Series with a Systematic Literature Review and Meta-Analysis. *Semin Ophthalmol* 2015;30(5-6):345-51.
11 Andres-Guerrero V, Perucho-Gonzalez L, Garcia-Feijoo J, Morales-Fernandez L, Saenz-Frances F, Herrero-Vanrell R, et al. Current Perspectives on the Use of Anti-VEGF Drugs as Adjuvant Therapy in Glaucoma. *Adv Ther* 2017;34(2):378-95.
12 Hwang HB, Han JW, Yim HB, Lee NY. Beneficial effects of adjuvant intravitreal bevacizumab injection on outcomes of Ahmed glaucoma valve implantation in patients with neovascular glaucoma: systematic literature review. *J Ocul Pharmacol Ther* 2015;31(4):198-203.
13 Zhou M, Xu X, Zhang X, Sun X. Clinical Outcomes of Ahmed Glaucoma Valve Implantation With or Without Intravitreal Bevacizumab Pretreatment for Neovascular Glaucoma: A Systematic Review and Meta-Analysis. *J Glaucoma* 2016;25(7):551-7.
14 Zhou R, Sun Y, Chen H, Sha S, He M, Wang W. Laser trabeculoplasty for open-angle glaucoma: a systematic review and network meta-analysis. *Am J Ophthalmol* 2020.
15 Dong Z, Gong J, Liao R, Xu S. Effectiveness of multiple therapeutic strategies in neovascular glaucoma patients: A PRISMA-compliant network meta-analysis. *Medicine* 2018;97(14):e9897.
16 European Glaucoma Society Terminology and Guidelines for Glaucoma, 4th Edition - Chapter 2: Classification and terminology. *Br J Ophthalmol* 2017;101(5):73-127.
17 Ma KT, Yang JY, Kim JH, Kim NR, Hong S, Lee ES, et al. Surgical results of ahmed valve implantation with intraoperative bevacizumab injection in patients with neovascular glaucoma. *J Glaucoma* 2012;21(5):331-6.
18 Mahdy RA, Nada WM, Fawzy KM, Alnashar HY, Almosalamy SM. Efficacy of intravitreal bevacizumab with panretinal photocoagulation followed by Ahmed valve implantation in neovascular glaucoma. *J Glaucoma* 2013;22(9):768-72.

For peer review only

19 Zhou MW, Wang W, Huang WB, Chen SD, Li XY, Gao XB, et al. Adjunctive with versus without intravitreal bevacizumab injection before ahmed glaucoma valve implantation in the treatment of neovascular glaucoma. *Chinese medical journal* 2013;126(8):1412-7.
20 Kang JY, Nam KY, Lee SJ, Lee SU. The effect of intravitreal bevacizumab injection before Ahmed valve implantation in patients with neovascular glaucoma. *Int Ophthalmol* 2014;34(4):793-9.
21 Arcieri ES, Paula JS, Jorge R, Barella KA, Arcieri RS, Secches DJ, et al. Efficacy and safety of intravitreal bevacizumab in eyes with neovascular glaucoma undergoing Ahmed glaucoma valve implantation: 2-year follow-up. *Acta ophthalmol* 2015;93(1):e1-6.
22 Kwon J, Sung KR. Effect of Preoperative Intravitreal Bevacizumab on the Surgical Outcome of Neovascular Glaucoma at Different Stages. *Journal of ophthalmology* 2017;2017:7672485.
23 Yue-Bing L, Xin L, Tai-Xiang L, Xiao-Ying H. Effect of ahmed glaucoma drainage valve implantation and leizumab on neovascularization glaucoma. *International eye science* 2019;19(11):1853-6.
24 Kong F, Ma X, Fan S, Lu J, Qin X, Zou J. Efficacy of intravitreal lucentis or conbercept injection combined with Ahmed glaucoma valve implantation for treatment of neovascular glaucoma. *Journal of Jilin University Medicine Edition* 2017;43(6):1237-42.
25 Yildirim N, Yalvac IS, Sahin A, Ozeer A, Bozca T. A Comparative Study Between Diode Laser Cyclophotocoagulation and the Ahmed Glaucoma Valve Implant in Neovascular Glaucoma A Long-term Follow-up. *J Glaucoma* 2009;18(3):192-6.
26 Liu XN, Tong FF. Study of surgical treatment on neovascular glaucoma. *International Eye Science* 2014;14(5):942-4.
27 Choy BNK, Lai JSM, Yeung JCC, Chan JCH. Randomized comparative trial of diode laser transscleral cyclophotocoagulation versus Ahmed glaucoma valve for neovascular glaucoma in Chinese – A pilot study. *Clin Ophthalmol* 2018;12:2545-52.
28 Zhang XL, Li SY, Zhang ZP, Miao PJ, Chen XY. Clinical effect of trabeculectomy and glaucoma valve implantation on neovascular glaucoma. *International Eye Science* 2014;14(2):349-51.
29 Sun JT, Liang HJ, An M, Wang DB. Efficacy and safety of intravitreal ranibizumab with panretinal photocoagulation followed by trabeculectomy compared with Ahmed glaucoma valve implantation in neovascular glaucoma. *International journal of ophthalmology* 2017;10(3):400-5.
30 Gao L, Liu W. Intravitreal ranibizumab injection combined trabeculectomy versus Ahmed glaucoma valve implantation in the treatment of neovascular glaucoma. *Zhonghua Shiyan Yanke Zazhi/Chinese Journal of Experimental Ophthalmology* 2018;36(8):613-8.
31 Du QW, Yang LS. Effects of intravitreal injection of Bevacizumab combined with complex trabeculectomy for advanced neovascular glaucoma. *International Eye Science* 2015;15(10):1766-8.
32 Wang F, Wang LL. Safety of anti-VEGF drugs with trabeculectomy for neovascular glaucoma. *International eye science* 2016;16(5):837-40.
33 Saito Y, Higashide T, Takeda H, Ohkubo S, Sugiyama K. Beneficial effects of preoperative intravitreal bevacizumab on trabeculectomy outcomes in neovascular glaucoma. *Acta ophthalmol* 2010;88(1):96-102.
34 Takihara Y, Inatani M, Kawaji T, Fukushima M, Iwao K, Iwao M, et al. Combined intravitreal bevacizumab and trabeculectomy with mitomycin C versus trabeculectomy with mitomycin C alone for neovascular glaucoma. *J Glaucoma* 2011;20(3):196-201.

For peer review only

35 Lee Y, Kim KS, Chang S. The effect of intravitreal bevacizumab injection before trabeculectomy in patients with neovascular glaucoma. *Invest Ophthalmol Vis Sci* 2015;56(7):2726.

36 Yu Z, Pu XL, Yan XN, Fan Q, Zhang M, Qin LL. Intravitreal injection of Conbercept combined with trabeculectomy and panretinal photocoagulation for the treatment of neovascular glaucoma. *International Eye Science* 2018;18(5):862-5.

37 Song RX. Effect of Conbercept Combined with Trabeculectomy on Aqueous Humor VEGF and SDF-1 Levels for Patients with Neovascular Glaucoma. *Chinese Journal of Pharmaceutical Biotechnology* 2019;26(6):521-4.

38 Shen CC, Salim S, Du H, Netland PA. Trabeculectomy versus ahmed glaucoma valve implantation in neovascular glaucoma. *Clin Ophthalmol* 2011;5(1):281-6.

39 Liu L, Xu Y, Huang Z, Wang X. Intravitreal ranibizumab injection combined trabeculectomy versus Ahmed valve surgery in the treatment of neovascular glaucoma: assessment of efficacy and complications. *BMC ophthalmology* 2016;16:65.

40 Mesquita J, Castro-de-Sousa JP, Vaz-Pereira S, Neves A, Passarinha LA, Tomaz CT. Vascular endothelial growth factors and placenta growth factor in retinal vasculopathies: Current research and future perspectives. *Cytokine Growth Factor Rev* 2018;39:102-15.

41 Riva I, Roberti G, Oddone F, Konstas AG, Quaranta L. Ahmed glaucoma valve implant: surgical technique and complications. *Clin Ophthalmol* 2017;11:357-67.

42 Suda M, Nakanishi H, Akagi T, Murakami T, Suzuma K, Suda K, et al. Baerveldt or ahmed glaucoma valve implantation with pars plana tube insertion in Japanese eyes with neovascular glaucoma: 1-year outcomes. *Clin Ophthalmol* 2018;12:2439-49.

43 Yalvac IS, Eksioglu U, Satana B, Duman S. Long-term results of Ahmed glaucoma valve and Molteno implant in neovascular glaucoma. *Eye* 2007;21(1):65-70.

44 SooHoo JR, Seibold LK, Kahook MY. Recent advances in the management of neovascular glaucoma. *Semin Ophthalmol* 2013;28(3):165-72.

45 Gedde SJ, Heuer DK, Parrish RK, 2nd, Tube Versus Trabeculectomy Study G. Review of results from the Tube Versus Trabeculectomy Study. *Curr Opin Ophthalmol* 2010;21(2):123-8.

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Figure 1. Flow chart of the study selection procedure.

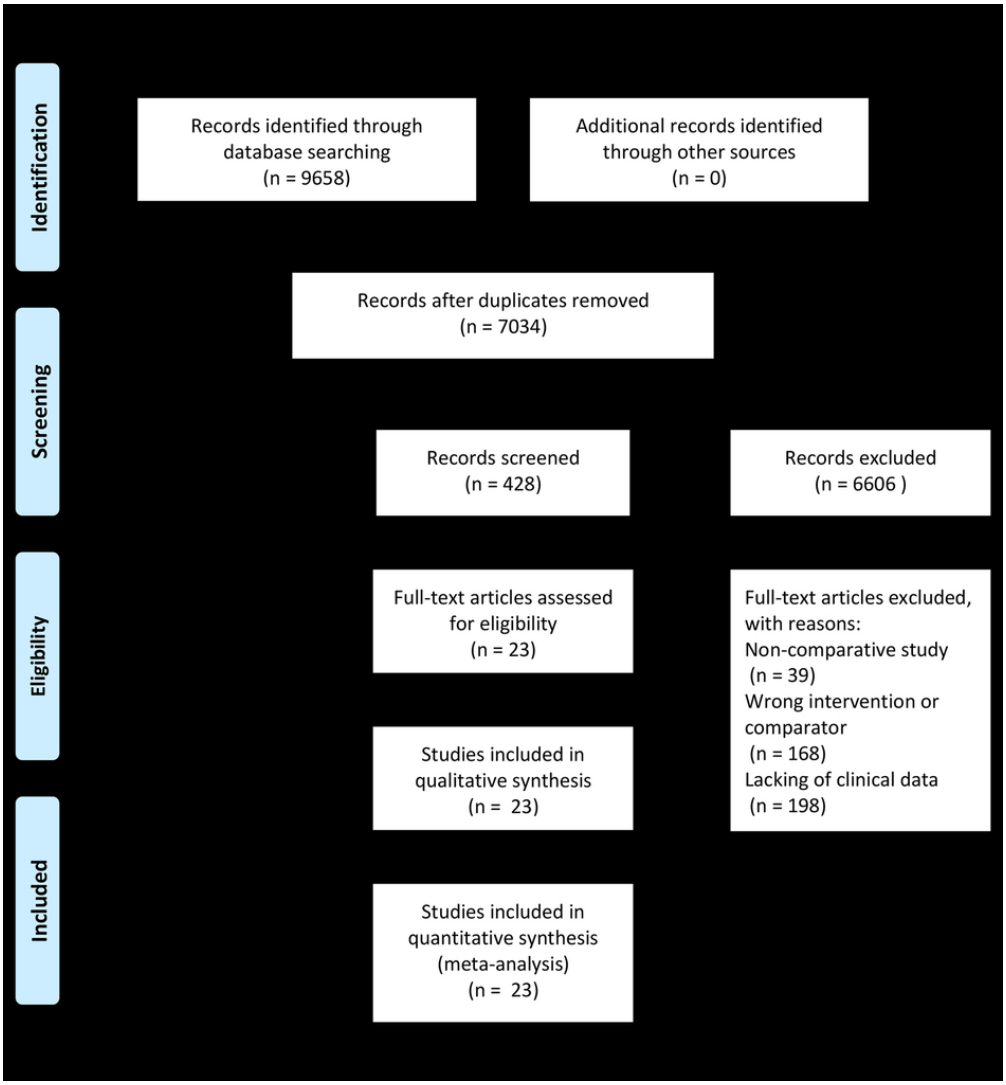
Figure 2. Network of treatment comparisons (Note: width of the lines is proportional to the number of trials comparing every pair of treatments. Size of each circle is proportional to the sample size of interventions. For example, in success rate, "9" represents the number of comparisons between AGV group and AGV+IVAV group, "n=257" represents the sample size of Trab(MMC)+IVAV group).

Figure 3. The results of network meta-analysis.

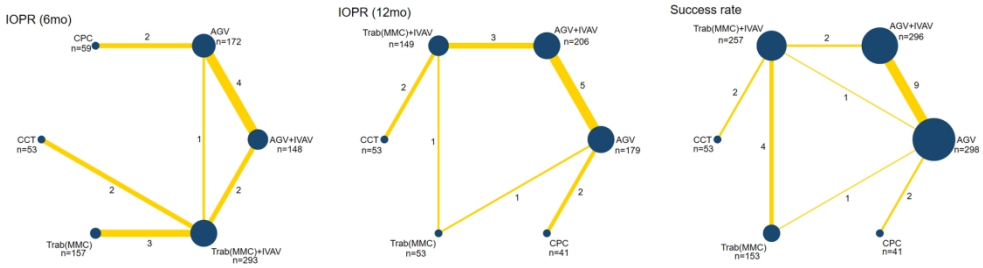
Figure 4. The contribution plots of each outcome (Note: 01=AGV, 02=AGV+IVAV, 03=CPC, 04=CCT, 05=Trab(MMC), and 06=Trab(MMC)+IVAV).

Figure 5. The funnel plots of each outcome (Note: 1=AGV, 2=AGV+IVAV, 3=CPC, 4=CCT, 5=Trab(MMC), and 6=Trab(MMC)+IVAV).

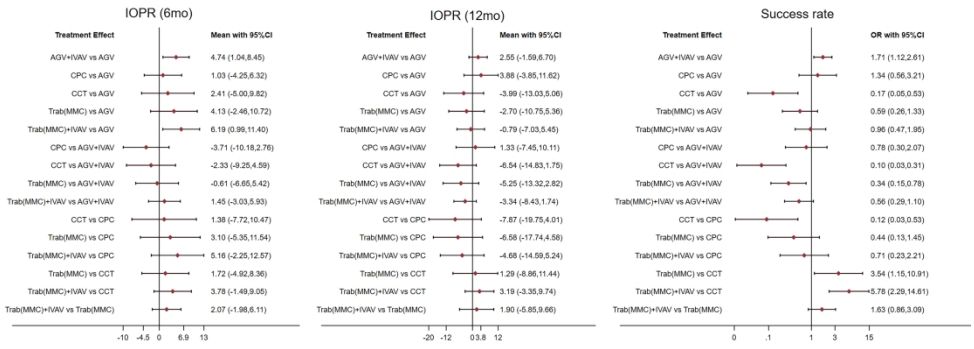
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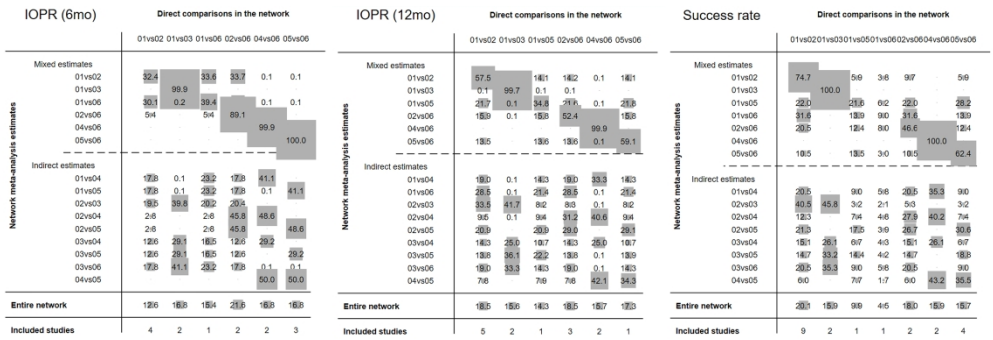
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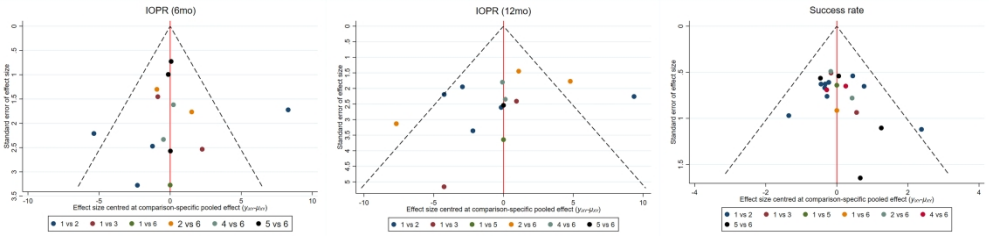
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Appendix 1. PubMed search strategy

1. neovascular Glaucoma as Topic [Mesh]
2. ((glaucoma* or angle* or iris or anterior) near/4 (neovascular*)) [tw]
3.((haemorrhagic or hemorrhagic or thrombotic or congestive or rubeotic or secondary) near/4 (glaucoma*)) [tw]
4. (NVG or NVI or NVA) [tw]
5. 1 or 2 or 3 or 4
6. randomized controlled trial [pt] or controlled clinical trial [pt] or randomized [tiab] or placebo [tiab] or clinical trials as topic [Mesh] or randomly [tiab] or trial [ti] or retrospective
7. animals [mh] not humans [mh]
8. 5 and 6 not 7

Appendix 2. Embase search strategy

1. neovascular glaucoma/exp
OR
2. ((glaucoma* or angle* or iris or anterior) adj4 neovascular*):ab
3. ((haemorrhagic or hemorrhagic or thrombotic or congestive or rubeotic or secondary) adj4 glaucoma*):ab
4. (NVG or NVI or NVA):ab
5. 1 or 2 or 3 or 4
6. [controlled clinical trial]/lim or [randomized controlled trial]/lim or retrospective
7. 5 and 6

Appendix 3. Central (Cochrane) search strategy

1. MeSH descriptor: [Glaucoma, Neovascular] explode all trees
2. (glaucoma* or angle* or iris or anterior) near/4 (neovascular*)
3. (haemorrhagic or hemorrhagic or thrombotic or congestive or rubeotic or secondary) near/4 (glaucoma*)
4. NVG or NVI or NVA
5. 1 or 2 or 3 or 4

Appendix 4. Web of Science search strategy

1. “Neovascular Glaucoma”
2. ((glaucoma* or angle* or iris or anterior) near/4 (neovascular*)) [tw]
3. ((haemorrhagic or hemorrhagic or thrombotic or congestive or rubeotic or secondary) near/4 (glaucoma*))[tw]
4. NVG or NVI or NVA
5. 1 or 2 or 3 or 4

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Appendix 5 Success and failure criteria applied in each study

References	Outcomes
Ma,2012	- A failure of operation was defined by: (1) IOP more than 21mmHg using a Goldmann application despite the use of antiglaucoma medication at 2 consecutive visits, (2) light perception-negative vision (except when preoperative vision acuity was light perception negative), (3) when additional glaucoma surgery was needed, or (4) operative or postoperative devastating complications, such as endopthalmitis, were noted.
Mahdy,2013	- Complete success was defined as an IOP≤21mmHg and ≥10mmHg without antiglaucoma medications or additional glaucoma surgery and without visually devastating complications such as endophthalmitis or phthisis bulbi and no loss of light perception. - Qualified success was considered when the above criteria were fulfilled, but with additions of antiglaucoma topical medical treatment. - Failure of shunt surgery in the study included instances when the above criteria of complete or qualified success were not fulfilled, a lack of control of IOP with or without glaucoma medications, operative or postoperative devastating complications, loss of light perception, or the need for additional glaucoma surgical intervention.
Zhou,2013	- Postoperative survival was defined as IOP 21mmHg, with or without glaucoma medications, and without significant vision-threatening complications (endophthalmitis, retinal detachment, suprachoroidal hemorrhage, preseptal cellulitis, or persistent hypotony (IOP 5mmHg)). - A complete success was defined as IOP between 6 and 21mmHg without any antiglaucoma medication. - Qualified success was defined as IOP between 6 and 21mmHg with one medication or more. - Failure was defined as IOP of more than 21mmHg despite the use of maximum-tolerated medications; IOP 5mmHg on two consecutive follow-up visits after 3 months; or the need for additional glaucoma surgery or laser to control IOP, removal of the shunt implant, or serious complication.
Kang,2014	- Surgical success was defined as an IOP≤21 mmHg with or without anti-glaucoma medication, and maintenance of visual acuity of light perception or better (except when preoperative visual acuity was NPL). - Surgical failure was defined as an IOP≥22 mmHg at two consecutive follow-up visits, or when additional glaucoma surgery was needed. Postoperative use of anti-glaucoma medication was not a criterion of success or failure.
Arcieri,2015	- Success was defined by the following criteria: (1) post-operative IOP level between 6 and 21 mmHg, with or without antiglaucoma medications, (2) IOP reduction of at least 30% relative to preoperative values. - Eyes requiring additional glaucoma surgery that developed phthisis or that showed loss of light perception were classified as failures.
Kwon,2017	- Surgical success was defined as an IOP between 6 and 21 mmHg, without loss of light perception (LP), and with or without the use of anti-glaucoma medication. - Surgical failure was defined as an IOP of more than 21 mmHg or less than 6 mmHg at two consecutive follow-up visits, the loss of LP, or a need for additional glaucoma interventions.

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Li,2019	- Complete success was defined as an IOP between 6 and 21 mmHg without the use of anti-glaucoma medication, and without serious ocular complication. - Qualified success was considered as an IOP between 6 and 21 mmHg, with 1 or 2 kinds of anti-glaucoma medication. - Failure was defined as: (1) IOP more than 21 mmHg or less than 6 mmHg, with 3 kinds of anti-glaucoma medication, (2) when additional glaucoma surgery was needed, (3) light perception-negative vision when preoperative vision acuity was light perception negative, (4) serious ocular complications were mentioned, or (5) removal of the shunt implant. - The total success rate was calculated as the sum of the complete success rate and the qualified success rate.
Kong,2017	- Complete success was defined as an IOP between 6 and 21 mmHg without the use of antiglaucoma medication, and without serious ocular complication. - Qualified success was defined as an IOP between 6 and 21 mmHg, with one medication or more. - Failure was defined as: (1) IOP more than 21 mmHg at two consecutive follow-up visits, with 3 kinds of anti-glaucoma medication or more, and a need for additional glaucoma surgery; (2) IOP less than 6 mmHg at consecutive follow-up visits; or (3) serious complications or (4) removal of the shunt implant due to ocular complications. - The total success rate was calculated as the sum of the complete success rate and the qualified success rate.
Yildirim,2009	- Surgical success was defined as IOP less than 21mmHg and greater than 5mm Hg without additional glaucoma surgery and without loss of light perception. Postoperative use of antiglaucoma medications was not accepted as a criterion of success or failure. The definition of hypotonia in this study was IOP of 5mm Hg or less in more than 2 consecutive visits. - The patients who underwent repeated DCPC and needling procedures were considered as treatment failures.
Choy,2018	- In the TSCP group, the laser procedure was repeated as necessary, spaced at least 4 weeks apart, until either the IOP was 21 mmHg or below, or a total of five treatments had been given. Uncontrolled IOP requiring any additional, non-assigned interventions was counted as failure. Additional procedures in relation to the implanted AGV (for example, clearing tube blockage, repair of implant exposure, excision of encapsulated bleb, and repositioning of implant) were not counted as failure but documented as complications. Similarly, any TSCP-related conjunctival burn or scleral thinning, with or without need for additional procedures, was recorded as a complication. - Successful IOP control was defined as a final visit IOP not higher than 21 mmHg, with or without medication, and no hypotony-related maculopathy or choroidal detachment. - Successful IOP control excluding the 1 TsCP and 5 agV eyes that did not follow the assigned protocol. - Overall success is defined as not only IOP normalization but also preservation or improvement of BCVA. Overall success = Valid cases – (failed IOP control + deteriorated BCVA).
Sun,2017	- Complete success was defined by the following criterion: postoperative IOP between 6 and 21 mm Hg without anti-glaucoma medications and without further anti-glaucoma surgery. - The qualified success criterion was IOP between 6 and 21 mmHg with one or more medications. - Failure was defined as IOP>21 mm Hg despite the use of maximum medications. If a patient maintained stable IOP after an additional surgery because of a complication, this was not considered failure of the procedure; however, the additional surgery could not be an

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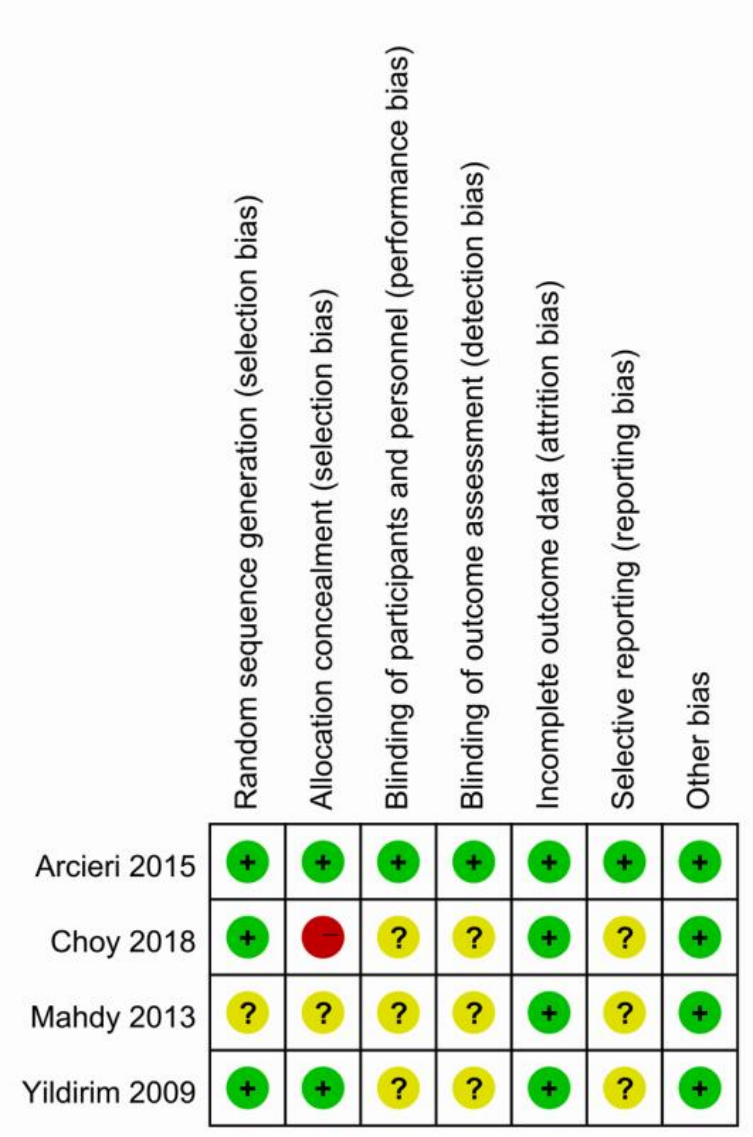
	anti-glaucoma surgery. - The total success rate was calculated as the sum of the complete success rate and the qualified success rate.
Gao,2018	- Complete success was defined as an IOP between 6 and 21 mmHg without the use of anti-glaucoma medication, and without serious ocular complication. - Qualified success was defined as an IOP between 6 and 21 mmHg, with one medication or more, and without serious ocular complication. - Failure was defined as: (1) IOP more than 21 mmHg at two consecutive follow-up visits, with 3 kinds of anti-glaucoma medication or more, and a need for additional glaucoma surgery; (2) IOP less than 6 mmHg at consecutive follow-up visits, with serious complications or (4) removal of the shunt implant due to ocular complications. - Serious complications included: (1) Drainage tube related complications: exposure, displacement, obstruction and corrosion of drainage tube; (2) Non-drainage tube related complications: retinal detachment, corneal decompensation, malignant glaucoma, supracoroidal hemorrhage, endophthalmitis, long-term low intraocular pressure. - The total success rate was calculated as the sum of the complete success rate and the qualified success rate.
Du,2015	- Surgical success was defined as an IOP between 7 and 22 mmHg, according to the general efficacy criteria.
Wang,2016	- Surgical success was defined as an IOP between 7 and 22 mmHg, according to the general efficacy criteria.
Sisto,2007	- Successes and failures were defined at the last post-operative visit. - Complete success was defined as post-operative IOP < 21 mmHg without medical therapy. - Qualified success was defined as post-operative IOP < 21 mmHg with medical therapy. - Failure was defined as postoperative IOP≥21 mmHg, despite maximal medical treatment, along with cases demonstrating no light perception(NLP) postoperatively.
Saito,2010	- The surgery was considered to have been successful when IOP < 21 mmHg with or without medication (qualified or complete success, respectively) was achieved. - A procedure was considered to have failed if IOP exceeded these criteria at two consecutive follow-up visits (follow-up visits were made monthly up to 1 year and once every several months thereafter) or in the presence of phthisis bulbi, loss of light perception or need for additional glaucoma surgeries, including surgical revision, trabeculectomy and cyclophotocoagulation.
Takihara,2011	- The surgical success and failure were defined before the data analysis. Surgical success was defined as an IOP ≤21 mmHg with or without topical ocular hypotensive medication and the maintenance of visual acuity of light perception or better. - Surgical failure was defined as an IOP continuously ≥22 mmHg at 2 consecutive follow-up visits, a deterioration of visual acuity to NPL, or additional glaucoma surgeries (such as filtering surgery and destruction of the ciliary body). We completed laser suture lysis within 2 months after MMCT. An IOP ≥22 mm Hg within 2 months after MMCT was not considered as a surgical failure because laser suture lysis was not completed.
Lee,2015	- Surgical success was defined as an IOP less than 21 mmHg, with or without anti-glaucoma medication and the maintenance of visual acuity of light perception or better. - Surgical failure was defined as more than 22 mmHg, a deterioration of visual acuity to NPL, or additional glaucoma surgeries (such as filtering surgery).

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Yu,2018	- Complete success was defined as an IOP less than 21 mmHg without anti-glaucoma medication. - Qualified success was defined as an IOP between 21 and 26 mmHg without anti-glaucoma medication, or IOP less than 21 mmHg with one medication. - Failure was defined as an IOP more than 30 mmHg, and still more than 21mmHg when using two kinds of medication.
Shen,2011	- Surgical success was defined as 6 mm Hg ≤ intraocular pressure ≤ 21 mm Hg, with or without glaucoma medications, without further glaucoma surgery including cyclophotocoagulation or complications that required removal of the Ahmed implant, and without loss of light perception. - For failure due to elevated intraocular pressure, two consecutive visits with intraocular pressure ≥ 21 mm Hg were required. Laser suture lysis and bleb needling to improve bleb function were not considered failure of the procedure.
Liu,2016	- Complete success was defined as IOP ≥ 6 mmHg and≤ 21 mmHg without any anti-glaucoma medications or further glaucoma surgery, and without loss of light perception. - Partial success was defined as IOP < 21 mmHg with topical anti-glaucoma medicines. - Surgical failure was defined as IOP ≥ 21 mmHg even with anti-glaucoma medicines, or additional urgical treatment was needed to control IOP , or loss of light perception.

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Appendix 6 Risk of bias summary of RCT



Note: The yellow circle with question mark represents “unclear risk of bias”, the red one with minus sign represents “high risk of bias” and the green one with plus sign represents “low risk of bias” (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

Appendix 7 The quality assessment according to the NOQAS of each cohort study

Study	Selection	Comparability	Outcome	Total score
Ma,2012	3	1	3	7
Zhou,2013	4	2	3	9
Kang,2014	3	2	2	7
Kwon,2017	3	2	3	8
Li,2019	3	2	2	7
Kong,2017	3	2	3	8
Liu,2014	3	2	1	6
Zhang,2014	3	2	1	6
Sun,2017	3	0	3	6
Gao,2018	3	2	3	8
Du,2015	3	0	3	6
Wang,2016	3	2	3	8
Saito,2010	3	2	3	8
Takihara,2011	3	2	3	8
Lee,2015	3	2	3	8
Yu,2018	4	2	2	8
Song,2019	4	2	1	7
Shen,2011	3	1	3	7
Liu,2016	3	1	2	6

Notes: NOQAS = New Castle – Ottawa Quality Assessment Scale.

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Appendix 8 Node-splitting analysis

Side/Outcomes	Direct		Indirect		Difference		P > |z|	tau
	Coefficient	Standard Error	Coefficient	Standard Error	Coefficient	Standard Error		
IOPR (6mo)								
A B	3.944	2.028	10.178	5.248	-6.234	5.624	0.268	3.241
A F	10.400	4.606	4.115	3.243	6.2847	5.633	0.265	3.241
B F	0.177	2.537	6.424	5.025	-6.247	5.628	0.267	3.241
D F	3.784	2.689	12.388	309.772	-8.604	309.783	0.978	3.251
E F	2.064	2.064	12.451	199.487	-10.387	199.497	0.958	3.251
IOPR (12mo)								
A B	2.550	2.316	2.605	8.314	-0.056	8.631	0.995	4.549
A E	-2.700	5.830	-2.754	6.405	0.054	8.661	0.995	4.551
B F	-3.405	2.917	-3.346	8.142	-0.059	8.649	0.995	4.550
D F	3.195	3.340	-1.584	330.177	4.779	330.193	0.988	4.241
E F	1.900	5.212	1.846	6.918	0.054	8.662	0.995	4.551
Success rate								
A B	0.564	0.227	0.256	0.707	0.308	0.743	0.679	0.000
A E	-0.205	0.641	-0.782	0.553	0.577	0.846	0.495	0.000
A F	-1.185	0.914	0.171	0.398	-1.356	0.997	0.174	0.000
B F	-0.483	0.415	-0.791	0.616	0.308	0.743	0.679	0.000
D F	1.754	0.473	-0.354	95.137	2.109	95.139	0.982	0.000
E F	0.595	0.359	0.018	0.766	0.577	0.846	0.495	0.000
Failure rate								
A B	-0.564	0.227	-0.117	0.744	-0.446	0.778	0.566	0.000
A E	0.205	0.641	0.848	0.566	-0.643	0.855	0.452	0.000
A F	2.197	1.143	-0.171	0.398	2.368	1.210	0.050	0.000
B F	0.483	0.415	0.930	0.658	-0.446	0.778	0.566	0.000
D F	-1.754	0.473	0.435	95.135	-2.189	95.137	0.982	0.000
E F	-0.595	0.359	0.049	0.776	-0.643	0.855	0.452	0.000

Notes: AGV, ahmed glaucoma valve implant surgery; CPC, cyclophotocoagulation; CCT, cyclocryotherapy; Trab, trabeculectomy; MMC, mitomycin; IVAV, intravitreal anti-VEGF; IOPR (6mo), intraocular pressure reduction 6 months after surgery; IOPR (12mo), intraocular pressure reduction 12 months after surgery; A=AGV, B=AGV+IVAV, C=CPC, D=CCT, E= Trab(MMC), F= Trab(MMC)+IVAV.

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Appendix 9 Pair-wise meta-analysis

Comparison

With statistically significant effect

Mean difference (95%CI) in IOP 6 months after surgery

AGV	4.74 (1.04,8.45)	1.03 (-4.25,6.32)	2.41 (-5.00,9.82)	4.13 (-2.46,10.72)	6.19 (0.99,11.40)
-2.55 (-6.70,1.59)	AGV+IVAV	-3.71 (-10.18,2.76)	-2.33 (-9.25,4.59)	-0.61 (-6.65,5.42)	1.45 (-3.03,5.93)
-3.88 (-11.62,3.85)	-1.33 (-10.11,7.45)	CPC	1.38 (-7.72,10.47)	3.10 (-5.35,11.54)	5.16 (-2.25,12.57)
3.99 (-5.06,13.03)	6.54 (-1.75,14.83)	7.87 (-4.01,19.75)	CCT	1.72 (-4.92,8.36)	3.78 (-1.49,9.05)
2.70 (-5.36,10.75)	5.25 (-2.82,13.32)	6.58 (-4.58,17.74)	-1.29 (-11.44,8.86)	Trab(MMC)	2.07 (-1.98,6.11)
0.79 (-5.45,7.03)	3.34 (-1.74,8.43)	4.68 (-5.24,14.59)	-3.19 (-9.74,3.35)	-1.90 (-9.66,5.85)	Trab(MMC) +IVAV

Mean difference (95%CI) in IOP 12 months after surgery

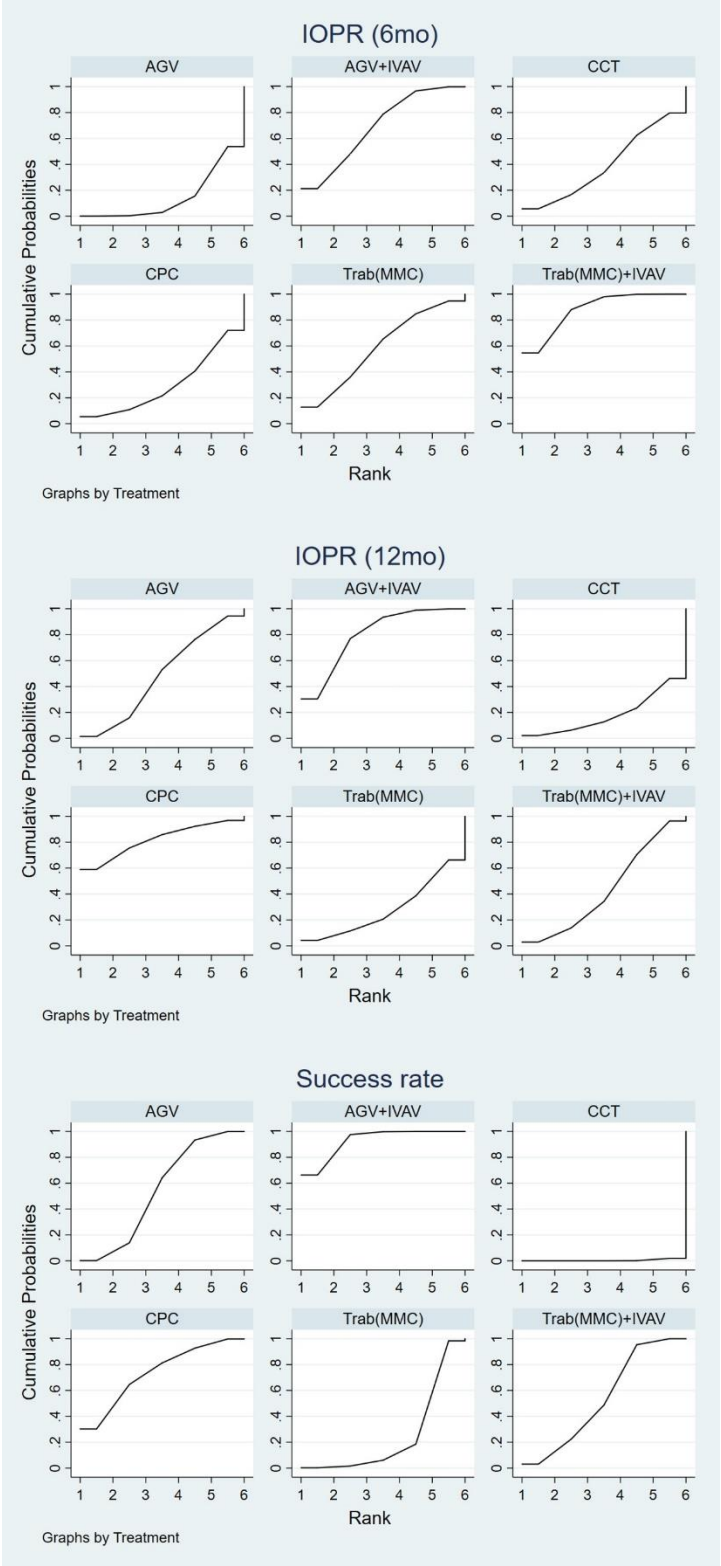
Odds ratio (95%CI) in failure rate

AGV	-0.53 (-0.95,-0.10)	-0.29 (-1.17,0.58)	1.84 (0.65,3.02)	0.57 (-0.27,1.40)	0.08 (-0.65,0.82)
-0.53 (-0.96,-0.11)	AGV+IVAV	0.23 (-0.74,1.21)	2.36 (1.21,3.52)	1.09 (0.27,1.92)	0.61 (-0.08,1.30)
-0.29 (-1.17,0.58)	0.24 (-0.73,1.21)	CPC	2.13 (0.66,3.60)	0.86 (-0.35,2.06)	0.38 (-0.77,1.52)
1.80 (0.63,2.97)	2.33 (1.19,3.48)	2.09 (0.63,3.55)	CCT	-1.27 (-2.40,-0.15)	-1.75 (-2.68,-0.83)
0.54 (-0.29,1.36)	1.07 (0.25,1.89)	0.83 (-0.37,2.03)	-1.26 (-2.39,-0.14)	Trab(MMC)	-0.48 (-1.12,0.16)
0.04 (-0.67,0.76)	0.58 (-0.10,1.25)	0.34 (-0.79,1.47)	-1.80 (-2.97,-0.63)	-0.49 (-1.13,0.15)	Trab(MMC) +IVAV

Odds ratio (95%CI) in success rate

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Appendix 10 The surface under the cumulative ranking plots based on cumulative probabilities of interventions



Appendix 11 GRADE for outcome measurements

Comparison	N (studies)	Within-study bias	Reporting bias	Indirectness	Imprecision	Heterogeneity	Incoherence	Confidence rating
IOPR (6mo)					Mixed evidence			
01vs 02	4	Some concerns	Undetected	No concerns	No concerns	Some concerns	No concerns	Moderate
01vs 03	2	Some concerns	Undetected	No concerns	No concerns	Some concerns	No concerns	Moderate
01vs 06	1	Some concerns	Undetected	No concerns	No concerns	Some concerns	No concerns	Moderate
02 vs 06	2	Some concerns	Undetected	No concerns	No concerns	Some concerns	No concerns	Moderate
04 vs 06	2	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
05 vs 06	3	Some concerns	Undetected	No concerns	No concerns	Some concerns	No concerns	Moderate
IOPR (6mo)					Indirect evidence			
01 vs 04	-	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
01 vs 05	-	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
02 vs 03	-	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
02 vs 04	-	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
02 vs 05	-	Some concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
03 vs 04	-	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
03 vs 05	-	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
03 vs 06	-	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
04 vs 05	-	Some concerns	Undetected	No concerns	No concerns	Some concerns	No concerns	Moderate
IOPR (12mo)					Mixed evidence			
01vs 02	5	Some concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
01vs 03	2	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
01 vs 05	1	Some concerns	Undetected	Some concerns	Some concerns	Some concerns	No concerns	Very low
02 vs 06	3	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
04 vs 06	2	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
05 vs 06	1	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
IOPR (12mo)					Indirect evidence			
01 vs 04	-	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
01 vs 06	-	Some concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
02 vs 03	-	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
02 vs 04	-	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
02 vs 05	-	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
03 vs 04	-	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
03 vs 05	-	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
03 vs 06	-	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
04 vs 05	-	Some concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low

Notes: N(studies)=number of studies, IOPR (6mo)= intraocular pressure reduction 6 months after surgery, IOPR (12mo)= intraocular pressure reduction 12 months after surgery; 01=AGV, 02=AGV+IVAV, 03=CPC, 04=CCT, 05=Trab(MMC), and 06=Trab(MMC)+IVAV.

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(Continue on the previous table)

Comparison	N (studies)	Within-study bias	Reporting bias	Indirectness	Imprecision	Heterogeneity	Incoherence	Confidence rating
Success rate				Mixed evidence				
01vs 02	9	Some concerns	Undetected	Some concerns	No concerns	No concerns	No concerns	Moderate
01vs 03	2	Some concerns	Undetected	Some concerns	Some concerns	No concerns	No concerns	Low
01 vs 05	1	Some concerns	Undetected	Some concerns	Some concerns	No concerns	No concerns	Low
01vs 06	1	Some concerns	Undetected	Some concerns	No concerns	No concerns	No concerns	Moderate
02 vs 06	2	Some concerns	Undetected	Some concerns	Some concerns	No concerns	No concerns	Low
04 vs 06	2	Some concerns	Undetected	Some concerns	No concerns	No concerns	No concerns	Moderate
05 vs 06	4	Some concerns	Undetected	Some concerns	No concerns	No concerns	No concerns	Moderate
Success rate				Indirect evidence				
01 vs 04	-	Some concerns	Undetected	Some concerns	No concerns	No concerns	No concerns	Moderate
02 vs 03	-	Some concerns	Undetected	Some concerns	Major concerns	No concerns	No concerns	Very low
02 vs 04	-	Some concerns	Undetected	Some concerns	No concerns	No concerns	No concerns	Moderate
02 vs 05	-	Some concerns	Undetected	Some concerns	No concerns	Some concerns	No concerns	Low
03 vs 04	-	Some concerns	Undetected	Some concerns	No concerns	No concerns	No concerns	Moderate
03 vs 05	-	Some concerns	Undetected	Some concerns	Some concerns	No concerns	No concerns	Low
03 vs 06	-	Some concerns	Undetected	Some concerns	Some concerns	Some concerns	No concerns	Very low
04 vs 05	-	Some concerns	Undetected	Some concerns	No concerns	No concerns	No concerns	Moderate

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Appendix 12 References list of included studies

1 . K. T. Ma, J. Y. Yang, J. H. Kim, et al. Surgical results of ahmed valve implantation with intraoperative bevacizumab injection in patients with neovascular glaucoma. Journal of Glaucoma 2012; 21(5): 331-336.
2 . R. A. Mahdy, W. M. Nada, K. M. Fawzy, et al. Efficacy of intravitreal bevacizumab with panretinal photocoagulation followed by Ahmed valve implantation in neovascular glaucoma. Journal of glaucoma 2013; 22(9): 768-772.
3 . M. W. Zhou, W. Wang, W. B. Huang, et al. Adjunctive with versus without intravitreal bevacizumab injection before ahmed glaucoma valve implantation in the treatment of neovascular glaucoma. Chinese Medical Journal 2013; 126(8): 1412-1417.
4 . J. Y. Kang, K. Y. Nam, S. J. Lee, et al. The effect of intravitreal bevacizumab injection before Ahmed valve implantation in patients with neovascular glaucoma. International Ophthalmology 2014; 34(4): 793-799.
5 . E. S. Arcieri, J. S. Paula, R. Jorge, et al. Efficacy and safety of intravitreal bevacizumab in eyes with neovascular glaucoma undergoing Ahmed glaucoma valve implantation: 2-year follow-up. Acta ophthalmologica 2015; 93(1): e1-6.
6 . J. Kwon,K. R. Sung. Effect of Preoperative Intravitreal Bevacizumab on the Surgical Outcome of Neovascular Glaucoma at Different Stages. J Ophthalmol 2017; 2017: 7672485.
7 . L. Yue-Bing, L. Xin, L. Tai-Xiang, et al. Effect of ahmed glaucoma drainage valve implantation and leizumab on neovascularization glaucoma. International eye science 2019; 19(11): 1853-1856.
8 . F. Kong, X. Ma, S. Fan, et al. Efficacy of intravitreal lucentis or conbercept injection combined with Ahmed glaucoma valve implantation for treatment of neovascular glaucoma. Journal of Jilin University Medicine Edition 2017; 43(6): 1237-1242.
9 . N. Yildirim, I. S. Yalvac, A. Sahin, et al. A Comparative Study Between Diode Laser Cyclophotocoagulation and the Ahmed Glaucoma Valve Implant in Neovascular Glaucoma A Long-term Follow-up. Journal of Glaucoma 2009; 18(3): 192-196.
10 . X. N. Liu,F. F. Tong. Study of surgical treatment on neovascular glaucoma. International Eye Science 2014; 14(5): 942-944.
11 . B. N. K. Choy, J. S. M. Lai, J. C. C. Yeung, et al. Randomized comparative trial of diode laser transscleral cyclophotocoagulation versus Ahmed glaucoma valve for neovascular glaucoma in Chinese – A pilot study. Clinical ophthalmology (Auckland, N.Z.) 2018; 12: 2545-2552.
12 . X. L. Zhang, S. Y. Li, Z. P. Zhang, et al. Clinical effect of trabeculectomy and glaucoma valve implantation on neovascular glaucoma. International Eye Science 2014; 14(2): 349-351.
13 . J. T. Sun, H. J. Liang, M. An, et al. Efficacy and safety of intravitreal ranibizumab with panretinal photocoagulation followed by trabeculectomy compared with Ahmed glaucoma valve implantation in neovascular glaucoma. Int J Ophthalmol 2017; 10(3): 400-405.
14 . L. Gao,W. Liu. Intravitreal ranibizumab injection combined trabeculectomy versus Ahmed glaucoma valve implantation in the treatment of neovascular glaucoma. Zhonghua Shiyan Yanke Zazhi/Chinese Journal of Experimental Ophthalmology 2018; 36(8): 613-618.
15 . Q. W. Du,L. S. Yang. Effects of intravitreal injection of Bevacizumab combined with complex trabeculectomy for advanced neovascular glaucoma. International Eye Science 2015; 15(10): 1766-1768.
16 . F. Wang,L. L. Wang. Safety of anti-VEGF drugs with trabeculectomy for neovascular glaucoma. International eye science 2016; 16(5): 837-840.
17 . Y. Saito, T. Higashide, H. Takeda, et al. Beneficial effects of preoperative intravitreal bevacizumab on trabeculectomy outcomes in neovascular glaucoma. Acta Ophthalmologica 2010; 88(1): 96-102.

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18 . Y. Takihara, M. Inatani, T. Kawaji, et al. Combined intravitreal bevacizumab and trabeculectomy with mitomycin C versus trabeculectomy with mitomycin C alone for neovascular glaucoma. Journal of Glaucoma 2011; 20(3): 196-201.
19 . Y. Lee, K. S. Kim,S. Chang. The effect of intravitreal bevacizumab injection before trabeculectomy in patients with neovascular glaucoma. Investigative Ophthalmology and Visual Science 2015; 56(7): 2726.
20 . Z. Yu, X. L. Pu, X. N. Yan, et al. Intravitreal injection of Conbercept combined with trabeculectomy and panretinal photocoagulation for the treatment of neovascular glaucoma. International Eye Science 2018; 18(5): 862-865.
21 . R. X. Song. Effect of Conbercept Combined with Trabeculectomy on Aqueous Humor VEGF and SDF-1 Levels for Patients with Neovascular Glaucoma. Chinese Journal of Pharmaceutical Biotechnology 2019; 26(6): 521-524.
22 . C. C. Shen, S. Salim, H. Du, et al. Trabeculectomy versus ahmed glaucoma valve implantation in neovascular glaucoma. Clinical Ophthalmology 2011; 5(1): 281-286.
23 . L. Liu, Y. Xu, Z. Huang, et al. Intravitreal ranibizumab injection combined trabeculectomy versus Ahmed valve surgery in the treatment of neovascular glaucoma: assessment of efficacy and complications. BMC ophthalmology 2016; 16: 65.

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	Page 1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	Page 3-4
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	Page 5-6
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	Page 6-7
METHODS			
Protocol and registration	5	The protocol is described in the Methods. Registration does not apply.	
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	Page 7-8
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	Page 7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Page 7 & Appendix
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Page 7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	Page 9
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	Page 9
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	Page 9
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	Page 8-9
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	Page 9-10

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	Page 9-10
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	Page 9-10
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Page 10 & Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Page 10-12
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Page 13 & Appendix
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Page 13-18
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Page 16
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Page 16-17 & Appendix
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Page 27
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	Page 18-20
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	Page 20-21
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	Page 21-22
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	Page 23

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Comparisons of the Short-term Effectiveness and Safety of Surgical Treatment for Neovascular Glaucoma: A Systematic Review and Network Meta-Analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-051794.R1
Article Type:	Original research
Date Submitted by the Author:	08-Feb-2022
Complete List of Authors:	Lin, Peijie; Shenzhen Eye Hospital Zhao, Qian; Shenzhen Eye Hospital He, Jing; Shenzhen Eye Hospital Fan, Wei; Jinan University, The 2rd College of Clinical Medicine He, Wenyi; Jinan University, The 2rd College of Clinical Medicine Lai, Mingying; Shenzhen Eye Hospital
Primary Subject Heading:	Ophthalmology
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Comparisons of the Short-term Effectiveness and Safety of Surgical Treatment for Neovascular Glaucoma: A Systematic Review and Network Meta-Analysis

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Running title: Network Meta-Analysis of Surgery for Neovascular Glaucoma

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Sources of support:

Supported by Science and Technology Plan Projects of Shenzhen, China

[JCYJ20160428144848002].

Word count: 4367 words.

Number of figures and tables: 5 figures and 5 tables.

Conflicts of interest:

All authors have completed the ICMJE uniform disclosure form. The authors have no conflicts of interest to declare.

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Abstract

Objective: To compare the effectiveness and safety of the six interventions for neovascular glaucoma.

Design: A systematic review and network meta-analysis.

Methods: Randomized controlled trials and cohort studies which compared the six interventions in neovascular glaucoma were identified using the following databases searched up to 1 September 2020: PubMed, Cochrane Library, Embase, Web of Science. The quality assessment was conducted by using the Cochrane risk of bias tool and the Newcastle-Ottawa scale. The primary outcome measure was the weighted mean differences (WMDs) for intraocular pressure reduction. Secondary one was odds ratios (ORs) for success rate. Outcome measures were reported with a 95% confidence interval (CI) and P< 0.05 was considered statistically significant. Network meta-analysis was performed using STATA version 15.0.

Results: Twenty-three studies involving a total of 1303 patients were included. The types of surgical treatments included ahmed glaucoma valve implant surgery (AGV), AGV combined with intravitreal anti-vascular endothelial growth factor (AGV+IVAV), cyclophotocoagulation (CPC), cyclocryotherapy (CCT), trabeculectomy with mitomycin (Trab(MMC)) and Trab(MMC) combined with IVAV (Trab(MMC)+IVAV). Network meta-analysis showed that in comparison with AGV, AGV+IVAV(MD=4.74, 95%CI 1.04 to 8.45) and Trab(MMC)+IVAV(MD=6.19, 95%CI 0.99 to 11.40) showed

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a favorable effect in intraocular pressure reduction (IOPR) 6 months after surgery. Compared with CCT, AGV (OR=-0.17, 95%CI -0.53 to -0.05), AGV+IVAV (OR=-0.10, 95%CI -3.48 to -1.19), CPC (OR=-0.12, 95%CI -0.53 to -0.05), Trab(MMC) (OR=3.54, 95%CI 1.15 to 10.91), Trab(MMC)+IVAV (OR=5.78, 95%CI 2.29 to 14.61) showed a superior impact in success rate. The order of efficacy as best intervention ranked as follows: Trab(MMC)+IVAV (IOPR 6 months after surgery, SUCRA=88.1), CPC (IOPR 12 months after surgery, SUCRA=81.9), AGV+IVAV (IOPR 12 months after surgery, SUCRA=79.9), AGV+IVAV (success rate, SUCRA=92.7). Adverse events were also summarized in detail.

Conclusion: In the treatment of neovascular glaucoma, AGV+IVAV and CPC were more effective in IOPR and success rate than the other four interventions**.** Additionally, AGV+IVAV is superior to CPC concerning the success rate in the long-term treatment. However, considering the limitations of this review, more high-quality trials, especially those surgical interventions not mentioned in this review, should be carried out in the future to further confirm the current findings.

Funding: Science and Technology Plan Projects of Shenzhen, China [JCYJ20160428144848002].

Trial registration and protocol: None.

Keywords: Ophthalmology; Medical ophthalmology; Glaucoma

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Article Summary: Strengths and limitations of this study

- ✓ To the best of our knowledge, this is the most comprehensive network meta-analysis which includes all the available data of comparative studies and evaluates different aspects of different therapeutic strategies.
- ✓ The effectiveness and the safety were assessed with rigorous inclusion criteria, leading to more convincing results.
- ✓ In terms of intraocular pressure reduction, we selected the common follow-up time point between all studies to reduce bias, including 6 months and 12 months.
- ✓ There was no common time point of data concerning success rate, we used information available at the final follow-up for statistical analysis.

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1 Introduction

Neovascular glaucoma (NVG) is a potentially blinding, refractory glaucoma, which is characterized by neovascularization (NV) in iris or anterior chamber angle.[1] NVG manifests as dramatic elevation of intraocular pressure (IOP), severe ocular pain and vision loss, influencing the quality of life in patients adversely. It is secondary to various ocular ischemic diseases, such as diabetic retinopathy, central retinal vein occlusion, and ocular ischemic syndrome.[2] With a high prevalence of diabetes and vascular disease, the incidence of NVG is increasing steadily, accounting for more than 30% of refractory glaucoma and becomes an occupational health issue.[1]

The therapy for NVG is two-fold. Firstly, the underlying cause needs controlling, usually by panretinal photocoagulation (PRP) or intravitreal vascular endothelial growth factor (VEGF) inhibitors, aiming at reduction of ischemic drive that induces formation of new blood vessels. The second key aspect is the successful IOP management. [3] When medication of decreasing IOP are insufficient, surgery is regarded as the first-line choice to for NVG, but complicated by higher failure rates and more difficult tissue anatomy than in primary glaucoma.[4] Thus, its treatments have been paid high attention to in glaucoma.

Currently, glaucoma drainage devices (GDDs),[5] filtering surgery,[6] cyclodestructive surgery,[7] combined with PRP or IVAV, are routinely utilized surgical treatment modalities for NVG,[8, 9] which present with a certain effectiveness and increase the diversity of treatment strategies.[10] Many authors verified the

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effectiveness of these treatment modalities.[11, 12] Nevertheless, there is no sufficient evidence of superiority of one over another. Some systematic review and meta-analyses have been published to evaluate the effectiveness and safety of these combination of treatment modalities for NVG. Two meta-analyses reported the effect of intravitreal bevacizumab injection before AGV implantation, which indicated higher surgical success rate.[13, 14] However, because of the limitations of the traditional pair-wise meta-analysis, it is difficult to determine which one is the best management of NVG in the modalities.

In recent years, network meta-analysis has been developed. Compared with traditional pairwise meta-analysis, it allows for combining of data related to multiple treatments, comparison of interventions based on indirect information, and generating a ranking of treatment arms according to the efficacy.[15] A latest network meta-analysis suggested that trabeculectomy with mitomycin and interferon was the most likely to improve the success rate in treatment of NVG,[16] followed by glaucoma valve; however, there was an inevitable insufficiency of evidence to determine the optimal surgical interventions since intraocular pressure reduction (IOPR) and adverse events were not mentioned in the analysis. The significant heterogeneity of follow-up time also indicated different degrees of bias in the analysis, which may affect the assessment of surgical success. In addition, the result of Loop-specific approach showed a certain local inconsistency signifying that the findings related to trabeculectomy should be interpreted cautiously.

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Therefore, aiming to add more sufficient evidence, we performed a network meta-analysis involving more specific outcome measurements including IOPR, success rate and adverse events to evaluate the comparative effectiveness and safety of different surgical treatment, to help ophthalmologist better make treatment strategies for patients with NVG. We present the following article in accordance with the PRISMA reporting checklist and MOOSE checklist.

2 Material and methods

2.1 Data Sources

A medical literature search was performed in the following databases from their inception through September 1st, 2020: PubMed, Web of Science, the Cochrane Library, Embase. Different from Dong's network meta-analysis,[16] we neglected the Chinese database due to the lower quality of literature retrieved. There are no restrictions regarding publication year, language, or methodological filter on these studies. The searching was performed using medical subject headings (MeSH) and keywords, including "neovascular glaucoma", "neovascular", "rubeosis and iris", "rubeosis iridis", "neovascularization and iris", "rubeotic glaucoma", "congestive and glaucoma", "haemorrhagic", "hemorrhagic", "NVG", "NVI" and "NVA". The details of search strategy are showed in Appendix 1 to 4 (available here). Two reviewers searched literatures, reviewed the titles and abstracts of articles independently to select the potential ones and tried contacting authors. The full texts of the selected articles were

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checked based on inclusion and exclusion criteria, and the divergent articles were checked by a third reviewer.

2.2 Inclusion Criteria

Studies were included according to the following criteria: (1) study type: randomized controlled trials (RCTs) or cohort studies; (2) population: patients were diagnosed as NVG by symptoms, signs and examination; (3) intervention: controlled study of different therapeutic strategies related to different surgeries; (4) outcome variables: at least one of the outcome measures was required.

2.3 Exclusion Criteria

The following studies were excluded if (1) follow-up time was < 6 months; (2) trials with a small sample size of (n < 10); (3) drug dose-related study; (4) comparative studies of similar surgical procedures, such as trabeculectomy versus modified trabeculectomy; (5) abstracts from conferences and full texts without raw data available for retrieval, literature reviews, letters and duplicate publications.

2.4 Outcome Measure

The primary outcome was the mean difference in IOPR after surgery, with or without anti-glaucoma medication. When authors reported the mean and SD of the IOPR, we used these directly. If not available, we computed them as follows: $IOPR = IOP_{baseline} - IOP_{endpoint}$, $SD_{IOPR} = (SD^2_{baseline} + SD^2_{endpoint} - SD^2_{baseline} \times SD^2_{endpoint})^{1/2}$. In order to reduce bias, we preferably selected the common follow-up time point between all

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studies: mean IOPR 6 months after surgery and mean IOPR 12 months after surgery. In case 12-month datas were not reported, we used the latest reported outcome data or the data closest to 12 months.[17] The secondary outcomes were the rates of surgical success, using the definitions used by authors of individual studies. In case there was no common time point of data concerning success rate, we used information available at the final follow-up for statistical analysis.

2.5 Data Extraction

Two investigators independently worked for data extraction, and they collected the following information: (1) basic characteristics, including author name, study design, publication year, age and gender of patients, intervention, sample size, stages of NVG, outcomes, and follow-up; (2) clinical outcomes, including IOPR 6 months after surgery, IOPR 12 months after surgery, and success rate.

2.6 Quality assessment

The quality assessment of RCTs was conducted by using the Cochrane risk of bias tool, and cohort studies using the Newcastle-Ottawa scale. Quality assessment was independently carried out by two reviewers, and disagreements were checked by a third reviewer.

2.7 Statistical analysis

A network meta-analysis was carried out using STATA version 15.0. Continuous variables (IOPR 6 months after surgery and IOPR 12 months after surgery) were

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analyzed using mean difference (MD) and its 95% credible interval (CI), while dichotomous variables (success rate) using Odds Ratio (OR). At the beginning of our network meta-analysis, pair-wise meta-analyses were performed, then "mvmeta" package was used to perform the plots of different comparisons, the rankplots based on probabilities and the surface under cumulative ranking (SUCRA) for different endpoints. Furthermore, node-splitting analysis and loop-specific approach were used to evaluate inconsistency, and the Grades of Recommendations Assessment, Development and Evaluation (GRADE) was used to evaluate the importance of the outcomes.

3 Results

3.1 Identification of the relevant studies

9658 articles were identified in our initial search, from which 2624 articles were excluded for duplications, and 6606 were excluded by reading titles and abstracts. In the remaining 428 articles, full texts were obtained to check eligibility, in which 39 studied were excluded because of non-comparative study, 168 studies were excluded for wrong intervention or comparator, and 198 studies were excluded because of lacking of clinical data. Finally, 23 studies were included in our final analysis (Appendix 5). Figure 1 shows the selection process for relevant studies.

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3.2 Characteristics of the included studies

The review involved 1303 patients and the sample size ranged from 20 to 170 cases. 4 studies were RCTs and 19 were cohort studies. 22 studies were two-arm studies, and 1 was three-arm study. In our study, the types of surgical treatments included AGV, AGV+IVAV, CPC, CCT, Trab(MMC) and Trab(MMC)+IVAV. To be more understandable, we explained the meaning of acronyms in Table 1. 8 studies compared AGV with AGV+IVAV, 3 studies compared AGV with CPC, 3 compared AGV+IVAV with Trab(MMC)+IVAV, 2 compared CCT with Trab(MMC)+IVAV, 5 compared Trab(MMC) with Trab(MMC)+IVAV, 1 compared AGV with Trab(MMC) and 1 compared AGV with Trab(MMC)+IVAV. 308 patients were included in AGV groups, 309 patients in AGV+IVAV groups, 67 patients in CPC groups, 53 patients in CCT groups, 235 patients in Trab(MMC) groups, and 355 patients in Trab(MMC)+IVAV groups. In terms of the criteria of IOPR, 14 trials employed the criteria of IOPR 6 months after surgery, 12 trials used the criteria of IOPR 12 months after surgery. As summarized in Appendix 6, 21 trials used heterogeneous criteria of success rate defined by authors of individual studies. The baseline characteristics of each study are presented in Table 2.

Table 1. The acronyms of the included surgical treatments

Acronym	Surgical treatment
AGV	Ahmed glaucoma valve implant surgery
IVAV	Intravitreal anti-vascular endothelial growth factor

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AGV+IVAV	AGV combined with IVAV
CPC	Cyclophotocoagulation
CCT	Cyclocryotherapy
Trab(MMC)	Trabeculectomy with mitomycin
Trab(MMC)+IVAV	Trab(MMC) combined with IVAV

Table 2. The characteristics of the included studies

References	Study design	Interventions	Age(MD,SD)	Gender (male/female)	Sample size		Outcomes	Mean follow-up(months)	anti-VEGF	Combine PRP
					No. Patients	No. Eyes				
Ma,2012[18]	OCT	AGV vs AGV+IVAV	58.78(13.09) vs 57.60(13.56)	27/25	N/A	32 vs 20	(c)(d)	15	Bevacizumab	Yes
Mahdy,2013[19]	RCT	AGV vs AGV+IVAV	56(4.3) vs 55(1.3)	23/17	20 vs 20	20 vs 20	(a)(b)(c)	18	Bevacizumab	Yes
Zhou,2013[20]	OCT	AGV vs AGV+IVAV	57.89(11.51) vs 54.40(13.68)	36/17	28 vs 25	28 vs 25	(a)(b)(c)	15.36(4.53) vs 15.08(5.14)	Bevacizumab	Yes
Kang,2014[21]	OCT	AGV vs AGV+IVAV	54.3(10.8) vs 54.8(13.0)	22/5	13 vs 14	13 vs 14	(c)	6	Bevacizumab	Yes
Arcieri,2015[22]	RCT	AGV vs AGV+IVAV	62.40(11.78) vs 59.25(8.05)	24/16	20 vs 20	20 vs 20	(a)(b)(c)	28.2(8.04) vs 25.8(8.04)	Bevacizumab	Yes
Kwon,2017[23]	OCT	AGV vs AGV+IVAV	57(18) vs 59(10)	56/14	25 vs 45	25 vs 45	(b)(c)	27(14) vs 26(16)	Bevacizumab	Yes
Li,2019[24]	OCT	AGV vs AGV+IVAV	57.49(8.42) vs 58.06(7.33)	46/34	34 vs 46	34 vs 46	(c)	6	Ranibizumab	N/A
Kong,2017[25]	OCT	AGV vs AGV+IVAV vs AGV+IVAV	54.24(11.05) vs 53.00(12.32) vs 54.00(12.74)	43/25	21 vs 26 vs 21	21 vs 26 vs 21	(a)(b)(c)	12	Ranibizumab; Conbercept	N/A
Yildirim,2009[26]	RCT	AGV vs CPC	57.2(10.3) vs 60.0(11.7)	35/31	33 vs 33	33 vs 33	(a)(b)(c)	24	No	N/A
Liu,2014[27]	OCT	AGV vs CPC	57.30(12.88) vs 56.44(13.16)	27/19	24 vs 22	31 vs 26	(a)	6	No	N/A
Choy,2018[28]	RCT	AGV vs CPC	62.8(11.0) vs 61.3(13.5)	13/6	12 vs 8	12 vs 8	(b)(c)	31.0(15.4) vs 28.5(17.9)	No	Yes
Zhang,2014[29]	OCT	AGV+IVAV vs Trab(MMC)+IVAV	57.5(12.5) vs 57.0(11.5)	12/10	12 vs 10	13 vs 10	(b)	26	Bevacizumab	Yes

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Sun,2017[30]	OCT	AGV+IVAV vs Trab(MMC)+IVAV	52.42(12.78)	23/22	23 vs 22	23 vs 22	(a)(b)(c)	12	Ranibizumab	Yes
Gao,2018[31]	OCT	AGV+IVAV vs Trab(MMC)+IVAV	56.25(9.98) vs 54.75(11.10)	49/27	36 vs 40	36 vs 40	(a)(b)(c)	12	Ranibizumab	Yes
Du,2015[32]	OCT	CCT vs Trab(MMC)+IVAV	57. 6(13. 4)	31/15	23 vs 23	23 vs 23	(a)(b)(c)	12	Bevacizumab	Yes
Wang,2016[33]	OCT	CCT vs Trab(MMC)+IVAV	45.5(4.6) vs 44.5(6.3)	33/27	30 vs 30	30 vs 30	(a)(b)(c)	12(0.3)	Ranibizumab	Yes
Saito,2010[34]	OCT	Trab(MMC) vs Trab(MMC)+IVAV	61(14) vs 60(13)	33/19	32 vs 20	32 vs 20	(c)	28(13) vs 12(5)	Bevacizumab	Yes
Takihara,2011[35]	OCT	Trab(MMC) vs Trab(MMC)+IVAV	60.0(10.4) vs 60.3(11.6)	44/6	30 vs 20	33 vs 24	(a)(b)(c)	25.3(9.8) vs 15.2(3.6)	Bevacizumab	Yes
Lee,2015[36]	OCT	Trab(MMC) vs Trab(MMC)+IVAV	52.73(11.27) vs 56.53(11.86)	38/20	26 vs 32	26 vs 32	(c)	12	Bevacizumab	Yes
Yu,2018[37]	OCT	Trab(MMC) vs Trab(MMC)+IVAV	54.88(4.19) vs 55.71(4.23)	48/42	42 vs 48	42 vs 48	(a)(c)	6	Conbercept	Yes
Song,2019[38]	OCT	Trab(MMC) vs Trab(MMC)+IVAV	59.25(5.51) vs 58.73(5.44)	91/79	82 vs 88	82 vs 88	(a)	6	Conbercept	N/A
Shen,2011[39]	OCT	AGV vs Trab(MMC)	54.0(15.6) vs 59.65(15.8)	19/21	20 vs 20	20 vs 20	(b)(c)	31.1(24.5) vs 25.0(19.7)	No	N/A
Liu,2016[40]	OCT	AGV vs Trab(MMC)+IVAV	56.7(13.6) vs 62.3(10.8)	21/16	19 vs 18	19 vs 18	(a)(c)	6	Ranibizumab	N/A

Notes: OCT, observational cohort study; RCT, randomized controlled trial; PRP, panretinal photocoagulation; AGV, ahmed glaucoma valve implant surgery; CPC, cyclophotocoagulation; CCT, cyclocryotherapy; MMC, mitomycin; Trab(MMC), trabeculectomy with MMC; VEGF, vascular endothelial growth factor; IVAV, intravitreal anti-VEGF; AGV+IVAV, AGV combined with IVAV; Trab(MMC)+IVAV, Trab(MMC) combined with IVAV;(a), intraocular pressure reduction 6 months after surgery;(b), intraocular pressure reduction 12 months after surgery;(c), success rate

3.3 Quality Assessment

In 4 RCTs, one studies didn't state the method of randomization in details, three study employed computer of random number for randomization. Only one study mentioned the blinding method. All the RCTs had complete data. The risk of bias summary is shown in Appendix 7. In terms of the cohort studies, based on the New Castle-Ottawa

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quality assessment scale, 1 study scored 9 points, 8 studies scored 8 points, 5 studies scored 7 points and 5 studies scored 6 points (Appendix 8).

3.4 The Results of Meta-Analysis

The results of pair-wise meta-analysis are demonstrated in Appendix 9. Figure 2 shows the network of eligible comparisons for IOPR 6 months after surgery, IOPR 12 months after surgery and success rate, and Figure 3 shows the results of network meta-analysis.

Fourteen studies involving 882 eyes reported IOPR 6 months after surgery (Figure 2). When compared with AGV, AGV+IVAV(MD=4.74, 95%CI 1.04 to 8.45) and Trab(MMC)+IVAV(MD=6.19, 95%CI 0.99 to 11.40) showed a significantly higher IOPR 6 months after surgery, but no significant difference was found between AGV+IVAV(MD=-1.45, 95%CI -5.93 to 3.03) and Trab(MMC)+IVAV. Meanwhile, no significant difference was found among the other interventions.

13 studies involving 681 eyes and 14 comparisons reported IOPR 12 months after surgery (Figure 2). As illustrated in Figure 3, no significant difference was found in IOPR 12 months after surgery.

In terms of success rate, 20 studies involving 21 comparisons and 1098 eyes were merged for analysis (Figure 2). Compared with CCT, the interventions including AGV(OR=-0.17, 95%CI -0.53 to -0.05), AGV+IVAV(OR=-0.10, 95%CI -3.48 to -1.19), CPC(OR=-0.12, 95%CI -0.53 to -0.05), Trab(MMC)(OR=3.54, 95%CI 1.15 to 10.91) and Trab(MMC)+IVAV(OR=5.78, 95%CI 2.29 to 14.61) presented with a

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significantly higher success rate. Additionally, the success rate in AGV+IVAV was significantly higher than that in AGV(OR=1.71, 95%CI 1.12 to 2.61) and Trab(MMC)(OR=-0.34, 95%CI -0.78 to -0.15), and no significant difference was found between the latter two treatments. However, no significant difference was found among the other interventions.

The plots of probability and SUCRA are illustrated in Appendix 10. Table 3 shows that Trab(MMC)+IVAV had the highest probability to be the best intervention in IOPR 6 months after surgery while AGV+IVAV ranking behind. CPC and AGV+IVAV had the highest probability to be the best intervention in IOPR 12 months after surgery, and AGV+IVAV had the highest probability to be the best intervention in success rate, respectively.

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Table 3. The results of SUCRA and probability.

Treatments/Outcomes	SUCRA	PrBest	MeanRank
IOPR (6mo)			
AGV	14.5	0.001	5.3
AGV+IVAV	68.9	0.213	2.6
CPC	30.1	0.054	4.5
CCT	39.6	0.057	4
Trab(MMC)	58.8	0.128	3.1
Trab(MMC)+IVAV	88.1	0.547	1.6
IOPR (12mo)			
AGV	48.2	0.014	3.6
AGV+IVAV	79.9	0.304	2
CPC	81.9	0.59	1.9
CCT	18.2	0.022	5.1
Trab(MMC)	28.2	0.042	4.6
Trab(MMC)+IVAV	43.6	0.029	3.8
Success rate			
AGV	54.3	0.002	3.3
AGV+IVAV	92.7	0.663	1.4
CPC	73.7	0.303	2.3
CCT	0.4	0	6
Trab(MMC)	25	0.002	4.8
Trab(MMC)+IVAV	53.9	0.03	3.3

Notes: data are probability in the rows of "SUCRA" and "PrBest". SUCRA= surface under the cumulative ranking, PrBest= the best probability (from 0 to 1), IOPR (6mo)= intraocular pressure reduction 6 months after surgery, IOPR (12mo)= intraocular pressure reduction 12 months after surgery.

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3.5 Consistency Analysis

Node-splitting analysis was performed to evaluate the inconsistency by comparing direct and indirect effects, indicating no significant inconsistency (Appendix 11) and the results were reliable. In addition, the results of Loop-specific approach showed no significant inconsistency in the comparisons of closed circles in outcomes of IOPR(12mo) and success rate, but significant inconsistency in IOPR(6mo) (Table 4).

Table 4. Loop-specific approach.

Outcome	Loop	ROR	Z_value	P_value	95%CI	Loop_Heterog_tau2
IOPR(6mo)	AGV-AGVI-TrabI	6.490	0.85	0.395	(0.00,21.45)	25.666
IOPR(12mo)	AGV-AGVI-Trab-TrabI	0.123	0.013	0.990	(0.00,19.05)	24.326
Success rate	AGV-Trab-TrabI	1.574	1.343	0.179	(0.00,3.87)	0
	AGV-AGVI-TrabI	1.282	1.16	0.246	(0.00,3.45)	0.066

Notes: loop-specific approach is used to check the inconsistency which aims at the closed loop. In this analysis, ROR is close to 1, indicating no significant difference between direct and indirect effects. AGVI= AGV+IVAV , Trab= Trab(MMC), TrabI= Trab(MMC)+IVAV.

3.6 GRADE for the Outcome Measurements

We summarized the GRADE judgements in Appendix 12. According to the suggestions of GRADE workgroups, we combined the evidences of direct and indirect comparisons and chose a higher level, and the results demonstrated the evidences provided in this review were moderate, low or very low.

3.7 Adverse Events

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In the included studies, 22 reported adverse events. Twenty studies reported 225 cases of hyphema, in which 70 occurred in the AGV group, 67 in the Trab(MMC) group, 38 in the AGV+IVAV group, 29 in the Trab(MMC)+IVAV group, 13 in the CCT group and 8 in the CPC group. Ten studies reported 73 cases of corneal oedema, in which 31 occurred in the CCT group, 20 in the AGV group, 14 in the CPC group, 5 in the AGV+IVAV group and 3 in the Trab(MMC) group. Eleven studies reported hypotony, in which 24 eyes occurred in the AGV+IVAV group, 17 in the Trab(MMC)+IVAV group, 16 in the AGV group, 7 in the Trab(MMC) group and 3 in the CPC group. Shallow anterior chamber was reported in 13 studies, including 42 cases in the AGV+IVAV group, 20 in the AGV group, 11 in the Trab(MMC)+IVAV group and 10 in the Trab(MMC) group.

Meanwhile, tube exposure or occlusion was reported in 8 studies, including 23 cases in the AGV group and 17 in the AGV+IVAV group. Encapsulated plate was reported in six studies, including 19 eyes in the AGV group and 12 in the AGV+IVAV group. Bleb leak was reported in six studies, including 6 cases in the Trab(MMC) group, 6 in the Trab(MMC)+IVAV group and 2 in the AGV+IVAV group. Phthisis bulbi was reported in four studies, including 6 eyes in the AGV group and 2 in the Trab(MMC) group.

Moreover, 5 studies reported 24 cases of vitreous hemorrage, in which 10 occurred in the Trab(MMC) group, 7 in the Trab(MMC)+IVAV group, 6 in the AGV group and 1 in the AGV+IVAV group. Choroidal effusion or detachment was reported in 11 studies, including 21 cases in the Trab(MMC)+ group, 18 in the Trab(MMC) group, 12 in the

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AGV group and 10 in the AGV+IVAV group. Two studies reported 2 cases with retinal detachment in the AGV group.

Besides, 9 studies reported anterior segment inflammation, 20 cases occurred in the AGV+IVAV group, 19 in the AGV group, 16 in the CCT group, 11 in the CPC group and 1 in the Trab(MMC)+IVAV group. Ocular pain was reported in five studies, in which 20 cases in the CCT group, 18 in the CPC group, 7 in the AGV group, 5 in the Trab(MMC) group, 2 in the AGV+IVAV group and 2 in the Trab(MMC)+IVAV group. Two studies reported ocular atrophy, 2 cases occurred in the CPC group and 1 in the Trab(MMC) group.

Regarding to adverse events, we supplemented extra data of complications (Appendix 13) and summarized them in Table 5.

Table 5. Adverse events

	AGV (n=213 eyes)	AGV+IVAV (n=269 eyes)	CPC (n=33 eyes)	CCT (n=53 eyes)	Trab(MMC) (n=182 eyes)	Trab(MMC)+IVAV (n=278 eyes)
Corneal edema	20 (9.4%)	5 (1.9%)	14 (42.4%)	31 (58.5%)	3 (1.6%)	—
Hyphema	70 (32.9%)	38 (14.1%)	8 (24.2%)	13 (24.5%)	68 (37.4%)	28 (10.1%)
Low IOP	16 (7.5%)	24 (8.9%)	3 (9.1%)	—	7 (3.8%)	17 (6.1%)
Shallow anterior	20 (9.4%)	40 (14.9%)	—	—	10 (5.5%)	11 (4.0%)
Vitreous hemorrhage	6 (2.8%)	1 (0.4%)	—	—	11 (6.0%)	6 (2.2%)
Choroidal detachment	12 (5.6%)	10 (3.7%)	—	—	18 (9.9%)	22 (7.9%)
Retinal detachment	2 (0.9%)	—	—	—	—	—
Tube exposure	12 (5.6%)	12 (4.5%)	—	—	—	—
Tube occlusion	13 (6.1%)	5 (1.9%)	—	—	—	—

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Encapsulated plate	19 (8.9%)	12 (4.5%)	—	—	—	—
Phthisis bulbi	6 (2.8%)	—	—	—	2 (1.1%)	—
Bleb leak	—	2 (0.7%)	—	—	6 (3.3%)	6 (2.2%)
Anterior segment inflammation	19 (8.9%)	20 (7.4%)	11 (33.3%)	16 (30.2%)	—	1 (0.4%)
Ocular pain	8 (3.8%)	2 (0.7%)	18 (54.5%)	36 (67.9%)	5 (2.7%)	2 (0.7%)
Ocular atrophy	—	—	2 (6.1%)	—	1 (0.5%)	—

Notes: AGV, ahmed glaucoma valve implant surgery; CPC, cyclophotocoagulation; CCT, cyclocryotherapy; MMC, mitomycin; Trab(MMC), trabeculectomy with MMC; VEGF, vascular endothelial growth factor; IVAV, intravitreal anti-VEGF; AGV+IVAV, AGV combined with IVAV; Trab(MMC)+IVAV, Trab(MMC) combined with IVAV.

4 Discussion

This is the first network meta-analysis to evaluate the efficacy and safety of the six widely utilized treatment modalities for NVG. The focus of our work is on more specific outcome measurements, with stricter inclusion criteria and less bias, which is very different from Dong's work. Firstly, we regrettably found that most of Dong's included studies were of poor quality as well as some extraction errors and failed to meet our inclusion criteria. Studies with follow-up time which was not mentioned or less than 6 months were excluded, and only two included studies in our meta-analysis were the same as Dong's.[19, 22] Meanwhile, since application of panretinal photocoagulation (PRP), laser peripheral iridotomy (LPI), and anti-scarring drugs were common as adjunctive treatments, they could not be compared as stand-alone interventions. Due to similar principles of reducing VEGF, bevacizumab, ranibizumab and conbercept were classified as anti-VEGF inhibitors in our analysis.[41] By reason

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of the foregoing, compared with Dong's network meta-analysis,[16] our inclusion criteria were more rigorous, leading to more convincing results.

The results demonstrated that in the six interventions, the effectiveness of CCT was the worst, and the effectiveness of AGV+IVAV was better than the other five interventions in success rat, but similar as CPC in IOPR 12 months after surgery. Besides, the effectiveness of Trab(MMC)+IVAV was the best among the six interventions in IOPR 6 months after surgery.

In terms of the rank of probability for the six interventions, it is reasonable that AGV+IVAV demonstrated a highest probability to be the best intervention in success rate. The insertion of GDD implants has been considered as an option with lower risk of failure than conventional filtering surgery for treating NVG.[5] Among GDDs, Ahmed possesses even higher acceptance and popularity and has been widely used by an increasing number of surgeons.[42] In this review, only AGV was included, we neglected other glaucoma valve-related studies in reason that there is no sufficient studies of comparison of each different types of GDDs. In regard to this, several authors conducted reviews and found no evidence of superiority of one over another in treatment of glaucoma.[5, 43, 44] Meanwhile, CPC and AGV+IVAV had an advantage over other four interventions in IOPR 12 months after surgery. The fact that two studies in the CPC groups were RCTs and patients with similar visual acuity were selected in the non-RCT, avoids create a potential selection bias, since cyclodestructive procedures are normally an option for advanced NVG with limited visual acuity, whereas AGVs

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are usually implanted in patients with better prognosis and visual potential.[45] Therefore, AGV+IVAV was superior to CPC concerning the success rate.

But it is noteworthy that Trab(MMC)+IVAV unexpectedly showed superiority in IOPR 6 months after surgery more than other five interventions. We think the reasons may be attributed to the following aspects. First of all, in this review most of trials have a small sample size, and the number of comparisons was small. As shown in Figure 4, the contribution plot of IOPR 6 months after surgery showed Trab(MMC)+IVAV accounted for a large proportion, demonstrating the small sample size influenced the total effect and final outcomes adversely (Figure 4). In addition, the funnel plots of IOPR 6 months after surgery showed the potential report bias (Figure 5). Moreover, the results of Loop-specific approach showed a significant inconsistency existed in IOPR 6 months after surgery, which means the results of indirect comparisons were not consistent with those of direct ones; two comparisons concluded different conclusions in IOPR 6 months after surgery. Subsequently, the findings in favor of Trab(MMC)+IVAV should be interpreted cautiously. Besides, in respect of IOPR 12 months after surgery and success rate, Trab(MMC)+IVAV showed inferiority to AGV+IVAV , CPC and AGV. As mainstream consideration, it shared a low long-term effectiveness mainly due to neovascular membrane obstruction in the filtering passage or external scarring and conjunctival fibrosis promoted by VEGF, even with antimetabolites such as mitomycin C or 5-fluorouracil.[6, 8]

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Moreover, as shown in Table 5, we had some interesting findings on adverse events. First, complications including corneal edema, anterior segment inflammation, ocular pain, and ocular atrophy were reported more in CPC and CCT cases than others. Second, hyphema was reported more in AGV, CPC, CCT and Trab(MMC) cases, whereas was less in AGV+IVAV and Trab(MMC)+IVAV cases, which indicated that combined surgery with IVAV may decrease the incidence of hyphema. Third, low IOP and shallow anterior seem to occur more in AGV+IVAV cases, may be due to the strong filterability of the implant at the beginning. Fourth, drainage tube related complications including tube exposure, tube occlusion and encapsulated plate occurred more often in AGV cases than those in AGV+IVAV cases, which demonstrated that anti-VEGF may improve surgical success rate of AGV. Additionally, bleb leak, vitreous hemorrhage and choroidal detachment were reported more in Trab(MMC) and Trab(MMC) +IVAV cases, and combined surgery with IVAV improved surgical effect; Last, it's worth noting that phthisis bulbi was reported in AGV and Trab(MMC) cases while retinal detachment was reported in AGV cases, although the incidence of these complications appeared low. However, there were some differences in the adverse events included in each study, the results need to be interpreted with caution.

Our review has two methodological strengths. In this research network meta-analysis was carried out to compare the direct and indirect effect of the four treatments, and the SUCRA plot was performed to estimate the ranks of interventions, which may facilitate ophthalmologist to make treatment strategies correctly. However, our review has its

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disadvantages. First, the evidence from GRADE for included outcomes was relatively low. Only four randomized studies were retrieved, whereas the remaining were non-randomized comparative studies. Second, there was a certain heterogeneity in follow-up time when observing 12-month postoperative IOP. We followed the recommendations of the European Glaucoma Society (European Glaucoma Society, 2017), which states that the latest reported outcome data or the data closest to 12 months could be reported in case 12-month data were not reported.[17] Third, heterogeneous definitions of the success criteria challenges the validity of our analysis and conclusions. Fourth, the number and sample size of retrieved studies are small, which may affect the interpretation of the results. Fifth, only three outcomes were analyzed in our research, more outcomes such as best corrected visual acuity (BCVA) and numbers of anti-glaucoma medications were also relevant but not analyzed, because no sufficient data was reported in the included studies. Sixth, since most of the primary ocular diseases leading to NVG were diabetic retinopathy and retinal vein occlusion (Appendix 14), which may do better than ocular ischemic syndrome cases or others, the results should be interpreted cautiously due to this potential confounding factor. Finally, the included studies only consisted of six interventions, more surgical treatment should be carried out in the upcoming studies to allow future systematic reviews and meta-analysis. These limitations may affect the final outcomes. In addition, data of each adverse event was insufficient, so the safety of the six treatments could not be evaluated by SUCRA in this review.

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5 Conclusion

In conclusion, our review suggested, among the six interventions, AGV+IVAV and CPC were superior to the other four interventions, and the effectiveness of CCT was the worst in the treatment of NVG. Additionally, AGV+IVAV is superior to CPC concerning the success rate in the long-term treatment. However, considering the limitations of this study, more high-quality trials, especially those surgical interventions not mentioned in this review, should be carried out in the future to further confirm the current conclusions.

Supplementary Materials

The supplementary appendix has 20 pages, including search strategy, references list of included studies, success and failure criteria applied in each study, risk of bias summary of RCT, the quality assessment according to the NOQAS of each OCT, node-splitting analysis, pair-wise meta-analysis, the surface under the cumulative ranking plots based on cumulative probabilities of interventions, GRADE for outcome measurements. More data can be found in the article. The interested reader can read a supplementary appendix online.

Data availability statement

Data are available in a public, open access repository. All data relevant to the study are included in the article or uploaded as online supplementary information. All included articles could be downloaded from the website https:// pubmed.ncbi.nlm.nih.gov/.

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Acknowledgments

We thank Haoyu Chen from Department of Ophthalmology, Joint Shantou International Eye Center Shantou University and the Chinese University of Hong Kong, for his editorial assistance.

Footnotes

Author contributions: PL conceived the study, drafted the protocol, collected data, performed the statistical analysis and wrote the manuscript. QZ participated in data extraction and data analysis. JH, WF and WH contributed to assembly of data, the quality assessment and data interpretation. ML reviewed and revised the manuscript. All authors have read and approved the final version of the manuscript.

Funding: This work was supported by Science and Technology Plan Projects of Shenzhen, China (grant number: JCYJ20160428144848002).

Competing interests: All authors have completed the ICMJE uniform disclosure form. The authors have no conflicts of interest to declare.

Financial disclose: The authors declare that they have no financial disclose.

Patient and Public Involvement: No patient involved.

Ethical approval and informed consent: Not applicable as no new patients were involved in this research.

Patient consent for publication: Not required.

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Provenance and peer review: Not commissioned; externally peer reviewed.

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References

1 Havens SJ, Gulati V. Neovascular Glaucoma. *Dev Ophthalmol* 2016;55:196-204.
2 Shazly TA, Latina MA. Neovascular glaucoma: etiology, diagnosis and prognosis. *Semin Ophthalmol* 2009;24(2):113-21.
3 Sivak-Callcott JA, O'Day DM, Gass JD, Tsai JC. Evidence-based recommendations for the diagnosis and treatment of neovascular glaucoma. *Ophthalmology* 2001;108(10):1767-76; quiz77, 800.
4 Hayreh SS. Neovascular glaucoma. *Prog Retin Eye Res* 2007;26(5):470-85.
5 Tseng VL, Coleman AL, Chang MY, Caprioli J. Aqueous shunts for glaucoma. *Cochrane Database Syst Rev* 2017;7:CD004918.
6 Dahan E, Ben Simon GJ. An augmented trabeculectomy for neovascular glaucoma. *Ophthalmic Surg Lasers Imaging* 2011;42(3):196-201.
7 Freigassner P, Eckhardt M. Transscleral cyclophotocoagulation versus cyclocryotherapy in treatment of neovascular glaucoma: a retrospective analysis. *Acta Ophthalmol Scand* 2003;81(6):674-5.
8 Yang H, Yu X, Sun X. Neovascular glaucoma: Handling in the future. *Taiwan J Ophthalmol* 2018;8(2):60-6.
9 Olmos LC, Lee RK. Medical and surgical treatment of neovascular glaucoma. *Int Ophthalmol Clin* 2011;51(3):27-36.
10 Andres-Guerrero V, Perucho-Gonzalez L, Garcia-Feijoo J, Morales-Fernandez L, Saenz-Frances F, Herrero-Vanrell R, et al. Current Perspectives on the Use of Anti-VEGF Drugs as Adjuvant Therapy in Glaucoma. *Adv Ther* 2017;34(2):378-95.
11 Shchomak Z, Cordeiro Sousa D, Leal I, Abegão Pinto L. Surgical treatment of neovascular glaucoma: a systematic review and meta-analysis. *Graefes Arch Clin Exp Ophthalmol* 2019;257(6):1079-89.
12 Lee NY, Hwang HB, Oh SH, Park CK. Efficacy of Additional Glaucoma Drainage Device Insertion in Refractory Glaucoma: Case Series with a Systematic Literature Review and Meta-Analysis. *Semin Ophthalmol* 2015;30(5-6):345-51.
13 Hwang HB, Han JW, Yim HB, Lee NY. Beneficial effects of adjuvant intravitreal bevacizumab injection on outcomes of Ahmed glaucoma valve implantation in patients with neovascular glaucoma: systematic literature review. *J Ocul Pharmacol Ther* 2015;31(4):198-203.
14 Zhou M, Xu X, Zhang X, Sun X. Clinical Outcomes of Ahmed Glaucoma Valve Implantation With or Without Intravitreal Bevacizumab Pretreatment for Neovascular Glaucoma: A Systematic Review and Meta-Analysis. *J Glaucoma* 2016;25(7):551-7.
15 Zhou R, Sun Y, Chen H, Sha S, He M, Wang W. Laser trabeculoplasty for open-angle glaucoma: a systematic review and network meta-analysis. *Am J Ophthalmol* 2020.
16 Dong Z, Gong J, Liao R, Xu S. Effectiveness of multiple therapeutic strategies in neovascular glaucoma patients: A PRISMA-compliant network meta-analysis. *Medicine* 2018;97(14):e9897.
17 European Glaucoma Society Terminology and Guidelines for Glaucoma, 4th Edition - Chapter 2: Classification and terminology. *Br J Ophthalmol* 2017;101(5):73-127.
18 Ma KT, Yang JY, Kim JH, Kim NR, Hong S, Lee ES, et al. Surgical results of ahmed valve implantation with intraoperative bevacizumab injection in patients with neovascular glaucoma. *J Glaucoma* 2012;21(5):331-6.

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19 Mahdy RA, Nada WM, Fawzy KM, Alnashar HY, Almosalamy SM. Efficacy of intravitreal bevacizumab with panretinal photocoagulation followed by Ahmed valve implantation in neovascular glaucoma. *J Glaucoma* 2013;22(9):768-72.
20 Zhou MW, Wang W, Huang WB, Chen SD, Li XY, Gao XB, et al. Adjunctive with versus without intravitreal bevacizumab injection before ahmed glaucoma valve implantation in the treatment of neovascular glaucoma. *Chinese medical journal* 2013;126(8):1412-7.
21 Kang JY, Nam KY, Lee SJ, Lee SU. The effect of intravitreal bevacizumab injection before Ahmed valve implantation in patients with neovascular glaucoma. *Int Ophthalmol* 2014;34(4):793-9.
22 Arcieri ES, Paula JS, Jorge R, Barella KA, Arcieri RS, Secches DJ, et al. Efficacy and safety of intravitreal bevacizumab in eyes with neovascular glaucoma undergoing Ahmed glaucoma valve implantation: 2-year follow-up. *Acta ophthalmologica* 2015;93(1):e1-6.
23 Kwon J, Sung KR. Effect of Preoperative Intravitreal Bevacizumab on the Surgical Outcome of Neovascular Glaucoma at Different Stages. *J Ophthalmol* 2017;2017:7672485.
24 Yue-Bing L, Xin L, Tai-Xiang L, Xiao-Ying H. Effect of ahmed glaucoma drainage valve implantation and leizumab on neovascularization glaucoma. *Int Eye Sci* 2019;19(11):1853-6.
25 Kong F, Ma X, Fan S, Lu J, Qin X, Zou J. Efficacy of intravitreal lucentis or conbercept injection combined with Ahmed glaucoma valve implantation for treatment of neovascular glaucoma. *Journal of Jilin University Medicine Edition* 2017;43(6):1237-42.
26 Yildirim N, Yalvac IS, Sahin A, Ozeer A, Bozca T. A Comparative Study Between Diode Laser Cyclophotocoagulation and the Ahmed Glaucoma Valve Implant in Neovascular Glaucoma A Long-term Follow-up. *J Glaucoma* 2009;18(3):192-6.
27 Liu XN, Tong FF. Study of surgical treatment on neovascular glaucoma. *Int Eye Sci* 2014;14(5):942-4.
28 Choy BNK, Lai JSM, Yeung JCC, Chan JCH. Randomized comparative trial of diode laser transscleral cyclophotocoagulation versus Ahmed glaucoma valve for neovascular glaucoma in Chinese – A pilot study. *Clin Ophthalmol* 2018;12:2545-52.
29 Zhang XL, Li SY, Zhang ZP, Miao PJ, Chen XY. Clinical effect of trabeculectomy and glaucoma valve implantation on neovascular glaucoma. *Int Eye Sci* 2014;14(2):349-51.
30 Sun JT, Liang HJ, An M, Wang DB. Efficacy and safety of intravitreal ranibizumab with panretinal photocoagulation followed by trabeculectomy compared with Ahmed glaucoma valve implantation in neovascular glaucoma. *International journal of ophthalmology* 2017;10(3):400-5.
31 Gao L, Liu W. Intravitreal ranibizumab injection combined trabeculectomy versus Ahmed glaucoma valve implantation in the treatment of neovascular glaucoma. *Zhonghua Shiyan Yanke Zazhi/Chinese Journal of Experimental Ophthalmology* 2018;36(8):613-8.
32 Du QW, Yang LS. Effects of intravitreal injection of Bevacizumab combined with complex trabeculectomy for advanced neovascular glaucoma. *Int Eye Sci* 2015;15(10):1766-8.
33 Wang F, Wang LL. Safety of anti-VEGF drugs with trabeculectomy for neovascular glaucoma. *Int Eye Sci* 2016;16(5):837-40.

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34 Saito Y, Higashide T, Takeda H, Ohkubo S, Sugiyama K. Beneficial effects of preoperative intravitreal bevacizumab on trabeculectomy outcomes in neovascular glaucoma. *Acta ophthalmologica* 2010;88(1):96-102.
35 Takihara Y, Inatani M, Kawaji T, Fukushima M, Iwao K, Iwao M, et al. Combined intravitreal bevacizumab and trabeculectomy with mitomycin C versus trabeculectomy with mitomycin C alone for neovascular glaucoma. *J Glaucoma* 2011;20(3):196-201.
36 Lee Y, Kim KS, Chang S. The effect of intravitreal bevacizumab injection before trabeculectomy in patients with neovascular glaucoma. *Invest Ophthalmol Vis Sci* 2015;56(7):2726.
37 Yu Z, Pu XL, Yan XN, Fan Q, Zhang M, Qin LL. Intravitreal injection of Conbercept combined with trabeculectomy and panretinal photocoagulation for the treatment of neovascular glaucoma. *Int Eye Sci* 2018;18(5):862-5.
38 Song RX. Effect of Conbercept Combined with Trabeculectomy on Aqueous Humor VEGF and SDF-1 Levels for Patients with Neovascular Glaucoma. *Chinese Journal of Pharmaceutical Biotechnology* 2019;26(6):521-4.
39 Shen CC, Salim S, Du H, Netland PA. Trabeculectomy versus ahmed glaucoma valve implantation in neovascular glaucoma. *Clin Ophthalmol* 2011;5(1):281-6.
40 Liu L, Xu Y, Huang Z, Wang X. Intravitreal ranibizumab injection combined trabeculectomy versus Ahmed valve surgery in the treatment of neovascular glaucoma: assessment of efficacy and complications. *BMC Ophthalmol* 2016;16:65.
41 Mesquita J, Castro-de-Sousa JP, Vaz-Pereira S, Neves A, Passarinha LA, Tomaz CT. Vascular endothelial growth factors and placenta growth factor in retinal vasculopathies: Current research and future perspectives. *Cytokine Growth Factor Rev* 2018;39:102-15.
42 Riva I, Roberti G, Oddone F, Konstas AG, Quaranta L. Ahmed glaucoma valve implant: surgical technique and complications. *Clin Ophthalmol* 2017;11:357-67.
43 Suda M, Nakanishi H, Akagi T, Murakami T, Suzuma K, Suda K, et al. Baerveldt or Ahmed glaucoma valve implantation with pars plana tube insertion in Japanese eyes with neovascular glaucoma: 1-year outcomes. *Clin Ophthalmol* 2018;12:2439-49.
44 Yalvac IS, Eksioglu U, Satana B, Duman S. Long-term results of Ahmed glaucoma valve and Molteno implant in neovascular glaucoma. *Eye* 2007;21(1):65-70.
45 SooHoo JR, Seibold LK, Kahook MY. Recent advances in the management of neovascular glaucoma. *Semin Ophthalmol* 2013;28(3):165-72.

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Figure 1. Flow chart of the study selection procedure.

Figure 2. Network of treatment comparisons (Note: width of the lines is proportional to the number of trials comparing every pair of treatments. Size of each circle is proportional to the sample size of interventions. For example, in success rate, "9" represents the number of comparisons between AGV group and AGV+IVAV group, "n=257" represents the sample size of Trab(MMC)+IVAV group).

Figure 3. The results of network meta-analysis.

Figure 4. The contribution plots of each outcome (Note: 01=AGV, 02=AGV+IVAV, 03=CPC, 04=CCT, 05=Trab(MMC), and 06=Trab(MMC)+IVAV).

Figure 5. The funnel plots of each outcome (Note: 1=AGV, 2=AGV+IVAV, 3=CPC, 4=CCT, 5=Trab(MMC), and 6=Trab(MMC)+IVAV).

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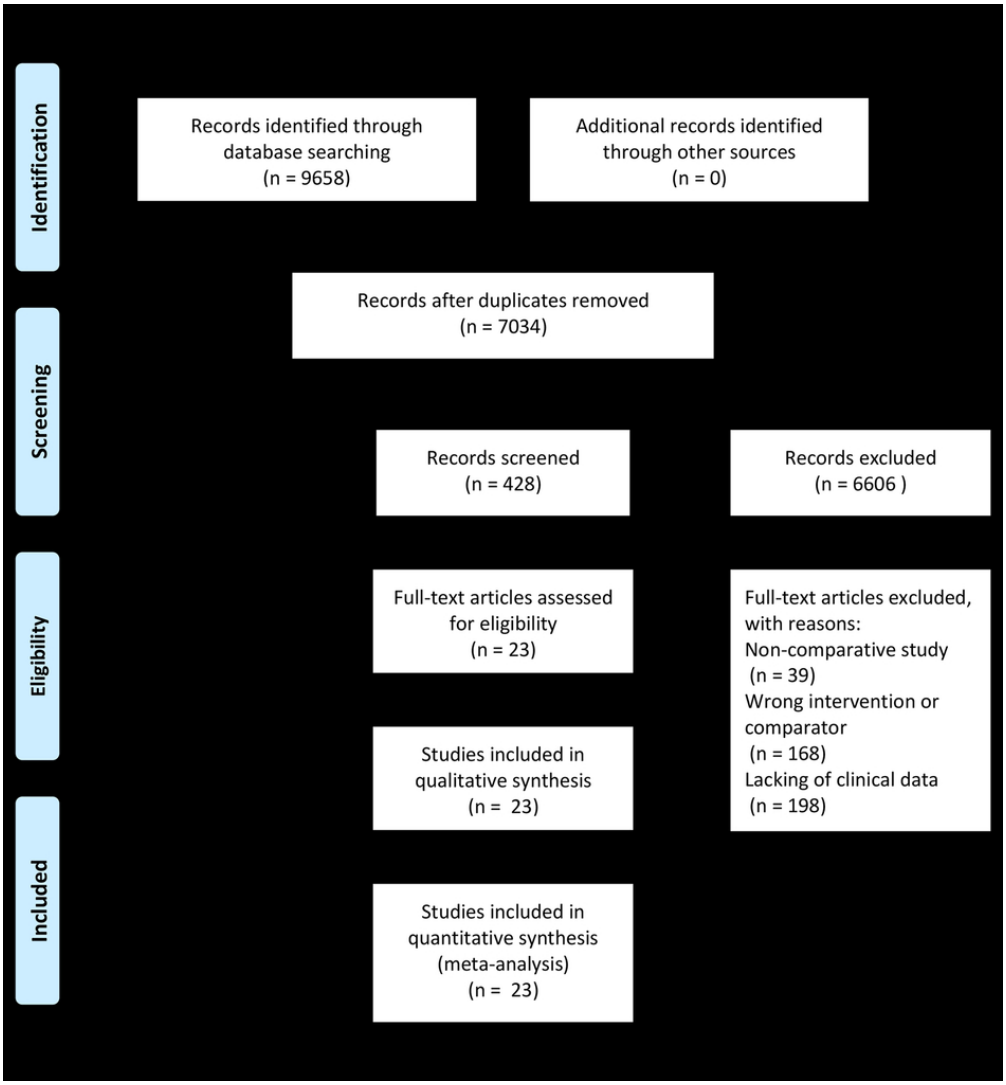


Figure 1. Flow chart of the study selection procedure.

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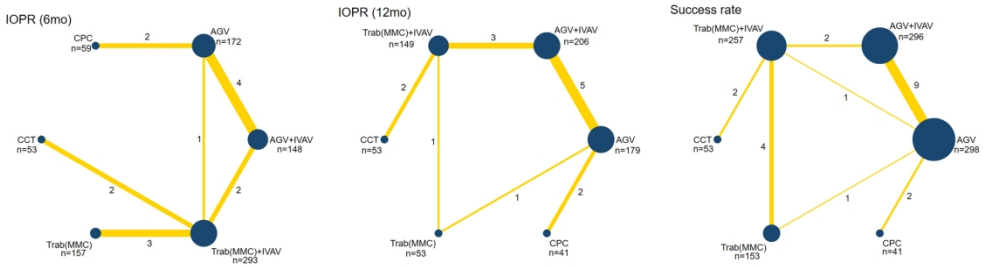


Figure 2. Network of treatment comparisons.

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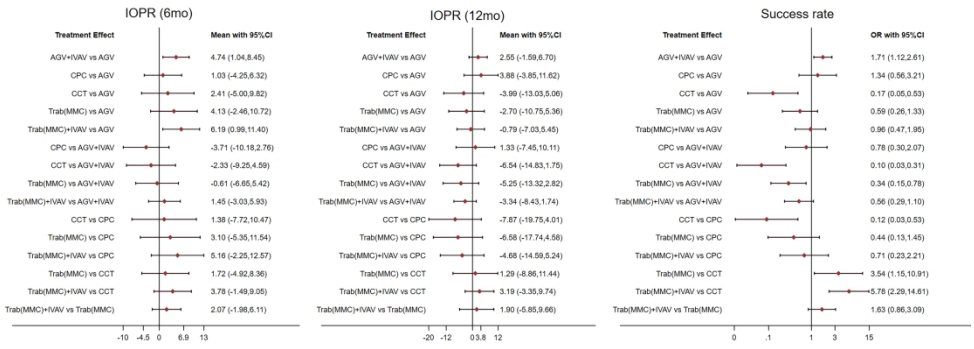


Figure 3. The results of network meta-analysis.

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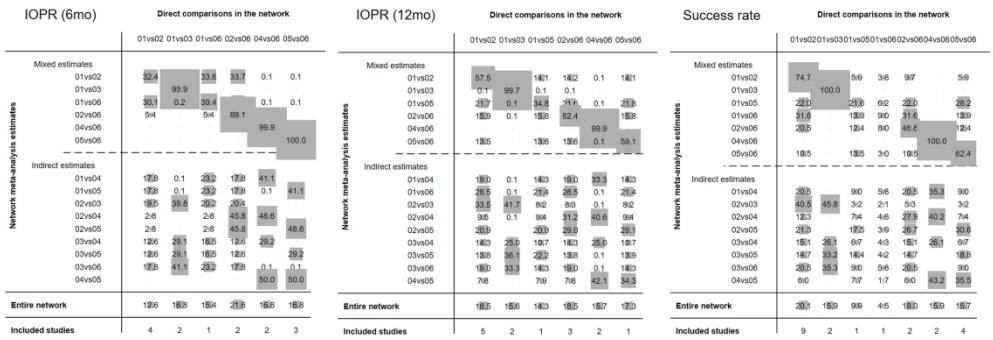


Figure 4. The contribution plots of each outcome.

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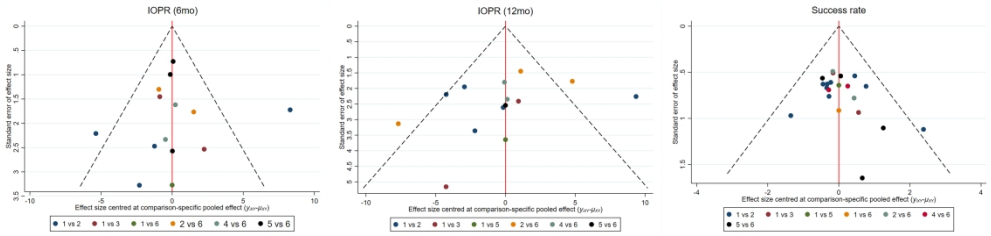


Figure 5. The funnel plots of each outcome.

178x43mm (600 x 600 DPI)

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Appendix 1. PubMed search strategy

1. neovascular Glaucoma as Topic [Mesh]
2. ((glaucoma* or angle* or iris or anterior) near/4 (neovascular*)) [tw]
3.((haemorrhagic or hemorrhagic or thrombotic or congestive or rubeotic or secondary) near/4 (glaucoma*)) [tw]
4. (NVG or NVI or NVA) [tw]
5. 1 or 2 or 3 or 4
6. randomized controlled trial [pt] or controlled clinical trial [pt] or randomized [tiab] or placebo [tiab] or clinical trials as topic [Mesh] or randomly [tiab] or trial [ti] or retrospective
7. animals [mh] not humans [mh]
8. 5 and 6 not 7

Appendix 2. Embase search strategy

1. neovascular glaucoma/exp
OR
2. ((glaucoma* or angle* or iris or anterior) adj4 neovascular*):ab
3. ((haemorrhagic or hemorrhagic or thrombotic or congestive or rubeotic or secondary) adj4 glaucoma*):ab
4. (NVG or NVI or NVA):ab
5. 1 or 2 or 3 or 4
6. [controlled clinical trial]/lim or [randomized controlled trial]/lim or retrospective
7. 5 and 6

Appendix 3. Central (Cochrane) search strategy

1. MeSH descriptor: [Glaucoma, Neovascular] explode all trees
2. (glaucoma* or angle* or iris or anterior) near/4 (neovascular*)
3. (haemorrhagic or hemorrhagic or thrombotic or congestive or rubeotic or secondary) near/4 (glaucoma*)
4. NVG or NVI or NVA
5. 1 or 2 or 3 or 4

Appendix 4. Web of Science search strategy

1. “Neovascular Glaucoma”
2. ((glaucoma* or angle* or iris or anterior) near/4 (neovascular*)) [tw]
3. ((haemorrhagic or hemorrhagic or thrombotic or congestive or rubeotic or secondary) near/4 (glaucoma*))[tw]
4. NVG or NVI or NVA
5. 1 or 2 or 3 or 4

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Appendix 5 References list of included studies

1 . K. T. Ma, J. Y. Yang, J. H. Kim, et al. Surgical results of ahmed valve implantation with intraoperative bevacizumab injection in patients with neovascular glaucoma. Journal of Glaucoma 2012; 21(5): 331-336.

2 . R. A. Mahdy, W. M. Nada, K. M. F Appendix 12 References list of included studies awzy, et al. Efficacy of intravitreal bevacizumab with panretinal photocoagulation followed by Ahmed valve implantation in neovascular glaucoma. Journal of glaucoma 2013; 22(9): 768-772.

3 . M. W. Zhou, W. Wang, W. B. Huang, et al. Adjunctive with versus without intravitreal bevacizumab injection before ahmed glaucoma valve implantation in the treatment of neovascular glaucoma. Chinese Medical Journal 2013; 126(8): 1412-1417.

4 . J. Y. Kang, K. Y. Nam, S. J. Lee, et al. The effect of intravitreal bevacizumab injection before Ahmed valve implantation in patients with neovascular glaucoma. International Ophthalmology 2014; 34(4): 793-799.

5 . E. S. Arcieri, J. S. Paula, R. Jorge, et al. Efficacy and safety of intravitreal bevacizumab in eyes with neovascular glaucoma undergoing Ahmed glaucoma valve implantation: 2-year follow-up. Acta ophthalmologica 2015; 93(1): e1-6.

6 . J. Kwon,K. R. Sung. Effect of Preoperative Intravitreal Bevacizumab on the Surgical Outcome of Neovascular Glaucoma at Different Stages. J Ophthalmol 2017; 2017: 7672485.

7 . L. Yue-Bing, L. Xin, L. Tai-Xiang, et al. Effect of ahmed glaucoma drainage valve implantation and leizumab on neovascularization glaucoma. International eye science 2019; 19(11): 1853-1856.

8 . F. Kong, X. Ma, S. Fan, et al. Efficacy of intravitreal lucentis or conbercept injection combined with Ahmed glaucoma valve implantation for treatment of neovascular glaucoma. Journal of Jilin University Medicine Edition 2017; 43(6): 1237-1242.

9 . N. Yildirim, I. S. Yalvac, A. Sahin, et al. A Comparative Study Between Diode Laser Cyclophotocoagulation and the Ahmed Glaucoma Valve Implant in Neovascular Glaucoma A Long-term Follow-up. Journal of Glaucoma 2009; 18(3): 192-196.

10 . X. N. Liu,F. F. Tong. Study of surgical treatment on neovascular glaucoma. International Eye Science 2014; 14(5): 942-944.

11 . B. N. K. Choy, J. S. M. Lai, J. C. C. Yeung, et al. Randomized comparative trial of diode laser transscleral cyclophotocoagulation versus Ahmed glaucoma valve for neovascular glaucoma in Chinese – A pilot study. Clinical ophthalmology (Auckland, N.Z.) 2018; 12: 2545-2552.

12 . X. L. Zhang, S. Y. Li, Z. P. Zhang, et al. Clinical effect of trabeculectomy and glaucoma valve implantation on neovascular glaucoma. International Eye Science 2014; 14(2): 349-351.

13 . J. T. Sun, H. J. Liang, M. An, et al. Efficacy and safety of intravitreal ranibizumab with panretinal photocoagulation followed by trabeculectomy compared with Ahmed glaucoma valve implantation in neovascular glaucoma. Int J Ophthalmol 2017; 10(3): 400-405.

14 . L. Gao,W. Liu. Intravitreal ranibizumab injection combined trabeculectomy versus Ahmed glaucoma valve implantation in the treatment of neovascular glaucoma. Zhonghua Shiyan Yanke Zazhi/Chinese Journal of Experimental Ophthalmology 2018; 36(8): 613-618.

15 . Q. W. Du,L. S. Yang. Effects of intravitreal injection of Bevacizumab combined with complex trabeculectomy for advanced neovascular glaucoma. International Eye Science 2015; 15(10): 1766-1768.

16 . F. Wang,L. L. Wang. Safety of anti-VEGF drugs with trabeculectomy for neovascular glaucoma. International eye science 2016; 16(5): 837-840.

17 . Y. Saito, T. Higashide, H. Takeda, et al. Beneficial effects of preoperative intravitreal bevacizumab on trabeculectomy outcomes in neovascular glaucoma. Acta Ophthalmologica 2010; 88(1): 96-102.

For peer review only

18 . Y. Takihara, M. Inatani, T. Kawaji, et al. Combined intravitreal bevacizumab and trabeculectomy with mitomycin C versus trabeculectomy with mitomycin C alone for neovascular glaucoma. Journal of Glaucoma 2011; 20(3): 196-201.

19 . Y. Lee, K. S. Kim,S. Chang. The effect of intravitreal bevacizumab injection before trabeculectomy in patients with neovascular glaucoma. Investigative Ophthalmology and Visual Science 2015; 56(7): 2726.

20 . Z. Yu, X. L. Pu, X. N. Yan, et al. Intravitreal injection of Conbercept combined with trabeculectomy and panretinal photocoagulation for the treatment of neovascular glaucoma. International Eye Science 2018; 18(5): 862-865.

21 . R. X. Song. Effect of Conbercept Combined with Trabeculectomy on Aqueous Humor VEGF and SDF-1 Levels for Patients with Neovascular Glaucoma. Chinese Journal of Pharmaceutical Biotechnology 2019; 26(6): 521-524.

22 . C. C. Shen, S. Salim, H. Du, et al. Trabeculectomy versus ahmed glaucoma valve implantation in neovascular glaucoma. Clinical Ophthalmology 2011; 5(1): 281-286.

23 . L. Liu, Y. Xu, Z. Huang, et al. Intravitreal ranibizumab injection combined trabeculectomy versus Ahmed valve surgery in the treatment of neovascular glaucoma: assessment of efficacy and complications. BMC ophthalmology 2016; 16: 65.

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Appendix 6 Success and failure criteria applied in each study

References	Outcomes
Ma,2012	- A failure of operation was defined by: (1) IOP more than 21mmHg using a Goldmann application despite the use of antiglaucoma medication at 2 consecutive visits, (2) light perception-negative vision (except when preoperative vision acuity was light perception negative), (3) when additional glaucoma surgery was needed, or (4) operative or postoperative devastating complications, such as endopthalmitis, were noted.
Mahdy,2013	- Complete success was defined as an IOP≤21mmHg and ≥10mmHg without antiglaucoma medications or additional glaucoma surgery and without visually devastating complications such as endophthalmitis or phthisis bulbi and no loss of light perception. - Qualified success was considered when the above criteria were fulfilled, but with additions of antiglaucoma topical medical treatment. - Failure of shunt surgery in the study included instances when the above criteria of complete or qualified success were not fulfilled, a lack of control of IOP with or without glaucoma medications, operative or postoperative devastating complications, loss of light perception, or the need for additional glaucoma surgical intervention.
Zhou,2013	- Postoperative survival was defined as IOP < 21mmHg, with or without glaucoma medications, and without significant vision-threatening complications (endophthalmitis, retinal detachment, suprachoroidal hemorrhage, preseptal cellulitis, or persistent hypotony (IOP < 5mmHg)). - A complete success was defined as IOP between 6 and 21mmHg without any antiglaucoma medication. - Qualified success was defined as IOP between 6 and 21mmHg with one medication or more. - Failure was defined as IOP of more than 21mmHg despite the use of maximum-tolerated medications; IOP < 5mmHg on two consecutive follow-up visits after 3 months; or the need for additional glaucoma surgery or laser to control IOP, removal of the shunt implant, or serious complication.
Kang,2014	- Surgical success was defined as an IOP≤21 mmHg with or without anti-glaucoma medication, and maintenance of visual acuity of light perception or better (except when preoperative visual acuity was NPL). - Surgical failure was defined as an IOP≥22 mmHg at two consecutive follow-up visits, or when additional glaucoma surgery was needed. Postoperative use of anti-glaucoma medication was not a criterion of success or failure.
Arcieri,2015	- Success was defined by the following criteria: (1) post-operative IOP level between 6 and 21 mmHg, with or without antiglaucoma medications, (2) IOP reduction of at least 30% relative to preoperative values. - Eyes requiring additional glaucoma surgery that developed phthisis or that showed loss of light perception were classified as failures.
Kwon,2017	- Surgical success was defined as an IOP between 6 and 21 mmHg, without loss of light perception (LP), and with or without the use of anti-glaucoma medication. - Surgical failure was defined as an IOP of more than 21 mmHg or less than 6 mmHg at two consecutive follow-up visits, the loss of LP, or a need for additional glaucoma interventions.

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Li,2019	- Complete success was defined as an IOP between 6 and 21 mmHg without the use of anti-glaucoma medication, and without serious ocular complication. - Qualified success was considered as an IOP between 6 and 21 mmHg, with 1 or 2 kinds of anti-glaucoma medication. - Failure was defined as: (1) IOP more than 21 mmHg or less than 6 mmHg, with 3 kinds of anti-glaucoma medication, (2) when additional glaucoma surgery was needed, (3) light perception-negative vision when preoperative vision acuity was light perception negative, (4) serious ocular complications were mentioned, or (5) removal of the shunt implant. - The total success rate was calculated as the sum of the complete success rate and the qualified success rate.
Kong,2017	- Complete success was defined as an IOP between 6 and 21 mmHg without the use of antiglaucoma medication, and without serious ocular complication. - Qualified success was defined as an IOP between 6 and 21 mmHg, with one medication or more. - Failure was defined as: (1) IOP more than 21 mmHg at two consecutive follow-up visits, with 3 kinds of anti-glaucoma medication or more, and a need for additional glaucoma surgery; (2) IOP less than 6 mmHg at consecutive follow-up visits; or (3) serious complications or (4) removal of the shunt implant due to ocular complications. - The total success rate was calculated as the sum of the complete success rate and the qualified success rate.
Yildirim,2009	- Surgical success was defined as IOP less than 21mmHg and greater than 5mm Hg without additional glaucoma surgery and without loss of light perception. Postoperative use of antiglaucoma medications was not accepted as a criterion of success or failure. The definition of hypotonia in this study was IOP of 5mm Hg or less in more than 2 consecutive visits. - The patients who underwent repeated DCPC and needling procedures were considered as treatment failures.
Choy,2018	- In the TSCP group, the laser procedure was repeated as necessary, spaced at least 4 weeks apart, until either the IOP was 21 mmHg or below, or a total of five treatments had been given. Uncontrolled IOP requiring any additional, non-assigned interventions was counted as failure. Additional procedures in relation to the implanted AGV (for example, clearing tube blockage, repair of implant exposure, excision of encapsulated bleb, and repositioning of implant) were not counted as failure but documented as complications. Similarly, any TSCP-related conjunctival burn or scleral thinning, with or without need for additional procedures, was recorded as a complication. - Successful IOP control was defined as a final visit IOP not higher than 21 mmHg, with or without medication, and no hypotony-related maculopathy or choroidal detachment. - Successful IOP control excluding the 1 TsCP and 5 agV eyes that did not follow the assigned protocol. - Overall success is defined as not only IOP normalization but also preservation or improvement of BCVA. Overall success = Valid cases – (failed IOP control + deteriorated BCVA).
Sun,2017	- Complete success was defined by the following criterion: postoperative IOP between 6 and 21 mm Hg without anti-glaucoma medications and without further anti-glaucoma surgery. - The qualified success criterion was IOP between 6 and 21 mmHg with one or more medications. - Failure was defined as IOP>21 mm Hg despite the use of maximum medications. If a patient maintained stable IOP after an additional surgery because of a complication, this was not considered failure of the procedure; however, the additional surgery could not be an

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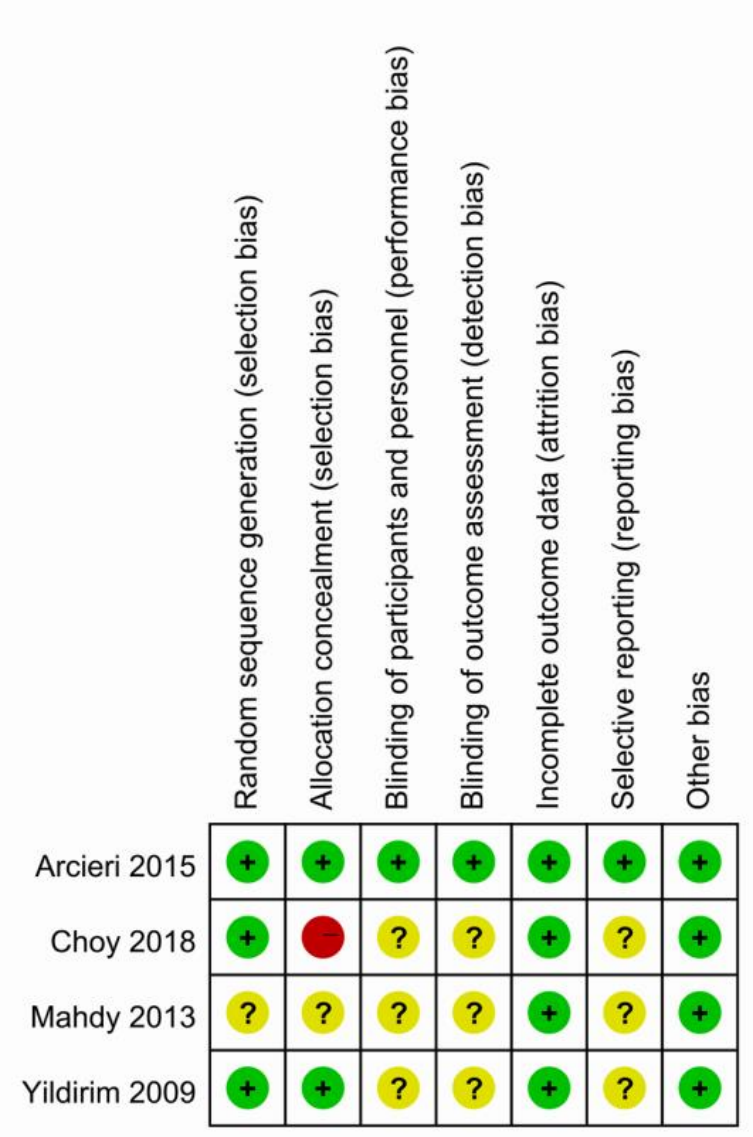
	anti-glaucoma surgery. - The total success rate was calculated as the sum of the complete success rate and the qualified success rate.
Gao,2018	- Complete success was defined as an IOP between 6 and 21 mmHg without the use of anti-glaucoma medication, and without serious ocular complication. - Qualified success was defined as an IOP between 6 and 21 mmHg, with one medication or more, and without serious ocular complication. - Failure was defined as: (1) IOP more than 21 mmHg at two consecutive follow-up visits, with 3 kinds of anti-glaucoma medication or more, and a need for additional glaucoma surgery; (2) IOP less than 6 mmHg at consecutive follow-up visits, with serious complications or (4) removal of the shunt implant due to ocular complications. - Serious complications included: (1) Drainage tube related complications: exposure, displacement, obstruction and corrosion of drainage tube; (2) Non-drainage tube related complications: retinal detachment, corneal decompensation, malignant glaucoma, supracoroidal hemorrhage, endophthalmitis, long-term low intraocular pressure. - The total success rate was calculated as the sum of the complete success rate and the qualified success rate.
Du,2015	- Surgical success was defined as an IOP between 7 and 22 mmHg, according to the general efficacy criteria.
Wang,2016	- Surgical success was defined as an IOP between 7 and 22 mmHg, according to the general efficacy criteria.
Sisto,2007	- Successes and failures were defined at the last post-operative visit. - Complete success was defined as post-operative IOP < 21 mmHg without medical therapy. - Qualified success was defined as post-operative IOP < 21 mmHg with medical therapy. - Failure was defined as postoperative IOP≥21 mmHg, despite maximal medical treatment, along with cases demonstrating no light perception(NLP) postoperatively.
Saito,2010	- The surgery was considered to have been successful when IOP < 21 mmHg with or without medication (qualified or complete success, respectively) was achieved. - A procedure was considered to have failed if IOP exceeded these criteria at two consecutive follow-up visits (follow-up visits were made monthly up to 1 year and once every several months thereafter) or in the presence of phthisis bulbi, loss of light perception or need for additional glaucoma surgeries, including surgical revision, trabeculectomy and cyclophotocoagulation.
Takihara,2011	- The surgical success and failure were defined before the data analysis. Surgical success was defined as an IOP ≤21 mmHg with or without topical ocular hypotensive medication and the maintenance of visual acuity of light perception or better. - Surgical failure was defined as an IOP continuously ≥22 mmHg at 2 consecutive follow-up visits, a deterioration of visual acuity to NPL, or additional glaucoma surgeries (such as filtering surgery and destruction of the ciliary body). We completed laser suture lysis within 2 months after MMCT. An IOP ≥22 mm Hg within 2 months after MMCT was not considered as a surgical failure because laser suture lysis was not completed.
Lee,2015	- Surgical success was defined as an IOP less than 21 mmHg, with or without anti-glaucoma medication and the maintenance of visual acuity of light perception or better. - Surgical failure was defined as more than 22 mmHg, a deterioration of visual acuity to NPL, or additional glaucoma surgeries (such as filtering surgery).

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Yu,2018	- Complete success was defined as an IOP less than 21 mmHg without anti-glaucoma medication. - Qualified success was defined as an IOP between 21 and 26 mmHg without anti-glaucoma medication, or IOP less than 21 mmHg with one medication. - Failure was defined as an IOP more than 30 mmHg, and still more than 21mmHg when using two kinds of medication.
Shen,2011	- Surgical success was defined as 6 mm Hg ≤ intraocular pressure ≤ 21 mm Hg, with or without glaucoma medications, without further glaucoma surgery including cyclophotocoagulation or complications that required removal of the Ahmed implant, and without loss of light perception. - For failure due to elevated intraocular pressure, two consecutive visits with intraocular pressure ≥ 21 mm Hg were required. Laser suture lysis and bleb needling to improve bleb function were not considered failure of the procedure.
Liu,2016	- Complete success was defined as IOP ≥ 6 mmHg and≤ 21 mmHg without any anti-glaucoma medications or further glaucoma surgery, and without loss of light perception. - Partial success was defined as IOP < 21 mmHg with topical anti-glaucoma medicines. - Surgical failure was defined as IOP ≥ 21 mmHg even with anti-glaucoma medicines, or additional urgical treatment was needed to control IOP , or loss of light perception.

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Appendix 7 Risk of bias summary of RCT



Note: The yellow circle with question mark represents “unclear risk of bias”, the red one with minus sign represents “high risk of bias” and the green one with plus sign represents “low risk of bias” (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

Notes: NOQAS = New Castle – Ottawa Quality Assessment Scale.

Appendix 8 The quality assessment according to the NOQAS of each cohort study

Study	Selection	Comparability	Outcome	Total score
Ma,2012	3	1	3	7
Zhou,2013	4	2	3	9
Kang,2014	3	2	2	7
Kwon,2017	3	2	3	8
Li,2019	3	2	2	7
Kong,2017	3	2	3	8
Liu,2014	3	2	1	6
Zhang,2014	3	2	1	6
Sun,2017	3	0	3	6
Gao,2018	3	2	3	8
Du,2015	3	0	3	6
Wang,2016	3	2	3	8
Saito,2010	3	2	3	8
Takihara,2011	3	2	3	8
Lee,2015	3	2	3	8
Yu,2018	4	2	2	8
Song,2019	4	2	1	7
Shen,2011	3	1	3	7
Liu,2016	3	1	2	6

Notes: NOQAS = New Castle – Ottawa Quality Assessment Scale.

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Appendix 9 Pair-wise meta-analysis

Comparison

With statistically significant effect

Mean difference (95%CI) in IOP 6 months after surgery

AGV	4.74 (1.04,8.45)	1.03 (-4.25,6.32)	2.41 (-5.00,9.82)	4.13 (-2.46,10.72)	6.19 (0.99,11.40)
-2.55 (-6.70,1.59)	AGV+IVAV	-3.71 (-10.18,2.76)	-2.33 (-9.25,4.59)	-0.61 (-6.65,5.42)	1.45 (-3.03,5.93)
-3.88 (-11.62,3.85)	-1.33 (-10.11,7.45)	CPC	1.38 (-7.72,10.47)	3.10 (-5.35,11.54)	5.16 (-2.25,12.57)
3.99 (-5.06,13.03)	6.54 (-1.75,14.83)	7.87 (-4.01,19.75)	CCT	1.72 (-4.92,8.36)	3.78 (-1.49,9.05)
2.70 (-5.36,10.75)	5.25 (-2.82,13.32)	6.58 (-4.58,17.74)	-1.29 (-11.44,8.86)	Trab(MMC)	2.07 (-1.98,6.11)
0.79 (-5.45,7.03)	3.34 (-1.74,8.43)	4.68 (-5.24,14.59)	-3.19 (-9.74,3.35)	-1.90 (-9.66,5.85)	Trab(MMC) +IVAV

Mean difference (95%CI) in IOP 12 months after surgery

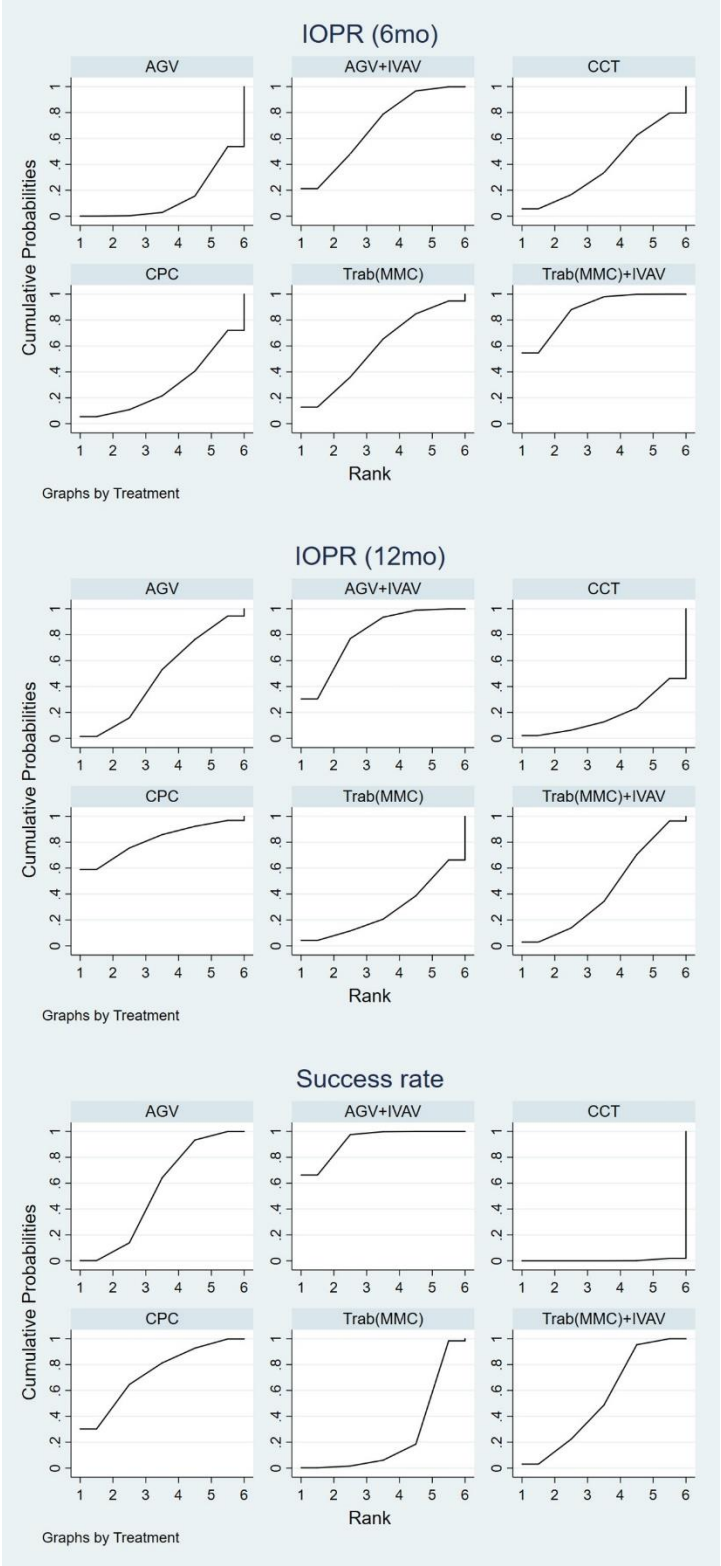
Odds ratio (95%CI) in failure rate

AGV	-0.53 (-0.95,-0.10)	-0.29 (-1.17,0.58)	1.84 (0.65,3.02)	0.57 (-0.27,1.40)	0.08 (-0.65,0.82)
-0.53 (-0.96,-0.11)	AGV+IVAV	0.23 (-0.74,1.21)	2.36 (1.21,3.52)	1.09 (0.27,1.92)	0.61 (-0.08,1.30)
-0.29 (-1.17,0.58)	0.24 (-0.73,1.21)	CPC	2.13 (0.66,3.60)	0.86 (-0.35,2.06)	0.38 (-0.77,1.52)
1.80 (0.63,2.97)	2.33 (1.19,3.48)	2.09 (0.63,3.55)	CCT	-1.27 (-2.40,-0.15)	-1.75 (-2.68,-0.83)
0.54 (-0.29,1.36)	1.07 (0.25,1.89)	0.83 (-0.37,2.03)	-1.26 (-2.39,-0.14)	Trab(MMC)	-0.48 (-1.12,0.16)
0.04 (-0.67,0.76)	0.58 (-0.10,1.25)	0.34 (-0.79,1.47)	-1.80 (-2.97,-0.63)	-0.49 (-1.13,0.15)	Trab(MMC) +IVAV

Odds ratio (95%CI) in success rate

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Appendix 10 The surface under the cumulative ranking plots based on cumulative probabilities of interventions



Appendix 11 Node-splitting analysis

Side/Outcomes	Direct		Indirect		Difference		P > |z|	tau
	Coefficient	Standard Error	Coefficient	Standard Error	Coefficient	Standard Error		
IOPR (6mo)								
A B	3.944	2.028	10.178	5.248	-6.234	5.624	0.268	3.241
A F	10.400	4.606	4.115	3.243	6.2847	5.633	0.265	3.241
B F	0.177	2.537	6.424	5.025	-6.247	5.628	0.267	3.241
D F	3.784	2.689	12.388	309.772	-8.604	309.783	0.978	3.251
E F	2.064	2.064	12.451	199.487	-10.387	199.497	0.958	3.251
IOPR (12mo)								
A B	2.550	2.316	2.605	8.314	-0.056	8.631	0.995	4.549
A E	-2.700	5.830	-2.754	6.405	0.054	8.661	0.995	4.551
B F	-3.405	2.917	-3.346	8.142	-0.059	8.649	0.995	4.550
D F	3.195	3.340	-1.584	330.177	4.779	330.193	0.988	4.241
E F	1.900	5.212	1.846	6.918	0.054	8.662	0.995	4.551
Success rate								
A B	0.564	0.227	0.256	0.707	0.308	0.743	0.679	0.000
A E	-0.205	0.641	-0.782	0.553	0.577	0.846	0.495	0.000
A F	-1.185	0.914	0.171	0.398	-1.356	0.997	0.174	0.000
B F	-0.483	0.415	-0.791	0.616	0.308	0.743	0.679	0.000
D F	1.754	0.473	-0.354	95.137	2.109	95.139	0.982	0.000
E F	0.595	0.359	0.018	0.766	0.577	0.846	0.495	0.000
Failure rate								
A B	-0.564	0.227	-0.117	0.744	-0.446	0.778	0.566	0.000
A E	0.205	0.641	0.848	0.566	-0.643	0.855	0.452	0.000
A F	2.197	1.143	-0.171	0.398	2.368	1.210	0.050	0.000
B F	0.483	0.415	0.930	0.658	-0.446	0.778	0.566	0.000
D F	-1.754	0.473	0.435	95.135	-2.189	95.137	0.982	0.000
E F	-0.595	0.359	0.049	0.776	-0.643	0.855	0.452	0.000

Notes: AGV, ahmed glaucoma valve implant surgery; CPC, cyclophotocoagulation; CCT, cyclocryotherapy; Trab, trabeculectomy; MMC, mitomycin; IVAV, intravitreal anti-VEGF; IOPR (6mo), intraocular pressure reduction 6 months after surgery; IOPR (12mo), intraocular pressure reduction 12 months after surgery; A=AGV, B=AGV+IVAV, C=CPC, D=CCT, E= Trab(MMC), F= Trab(MMC)+IVAV.

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Appendix 12 GRADE for outcome measurements

Comparison	N (studies)	Within-study bias	Reporting bias	Indirectness	Imprecision	Heterogeneity	Incoherence	Confidence rating
IOPR (6mo)					Mixed evidence			
01vs 02	4	Some concerns	Undetected	No concerns	No concerns	Some concerns	No concerns	Moderate
01vs 03	2	Some concerns	Undetected	No concerns	No concerns	Some concerns	No concerns	Moderate
01vs 06	1	Some concerns	Undetected	No concerns	No concerns	Some concerns	No concerns	Moderate
02 vs 06	2	Some concerns	Undetected	No concerns	No concerns	Some concerns	No concerns	Moderate
04 vs 06	2	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
05 vs 06	3	Some concerns	Undetected	No concerns	No concerns	Some concerns	No concerns	Moderate
IOPR (6mo)					Indirect evidence			
01 vs 04	-	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
01 vs 05	-	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
02 vs 03	-	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
02 vs 04	-	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
02 vs 05	-	Some concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
03 vs 04	-	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
03 vs 05	-	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
03 vs 06	-	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
04 vs 05	-	Some concerns	Undetected	No concerns	No concerns	Some concerns	No concerns	Moderate
IOPR (12mo)					Mixed evidence			
01vs 02	5	Some concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
01vs 03	2	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
01 vs 05	1	Some concerns	Undetected	Some concerns	Some concerns	Some concerns	No concerns	Very low
02 vs 06	3	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
04 vs 06	2	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
05 vs 06	1	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
IOPR (12mo)					Indirect evidence			
01 vs 04	-	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
01 vs 06	-	Some concerns	Undetected	No concerns	No concerns	Major concerns	No concerns	Low
02 vs 03	-	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
02 vs 04	-	Some concerns	Undetected	No concerns	Some concerns	No concerns	No concerns	Moderate
02 vs 05	-	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
03 vs 04	-	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
03 vs 05	-	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
03 vs 06	-	Some concerns	Undetected	No concerns	Some concerns	Some concerns	No concerns	Low
04 vs 05	-	Some concerns	Undetected	No concerns	Major concerns	No concerns	No concerns	Low

Notes: N(studies)=number of studies, IOPR (6mo)= intraocular pressure reduction 6 months after surgery, IOPR (12mo)= intraocular pressure reduction 12 months after surgery; 01=AGV, 02=AGV+IVAV, 03=CPC, 04=CCT, 05=Trab(MMC), and 06=Trab(MMC)+IVAV.

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(Continue on the previous table)

Comparison	N (studies)	Within-study bias	Reporting bias	Indirectness	Imprecision	Heterogeneity	Incoherence	Confidence rating
Success rate				Mixed evidence				
01vs 02	9	Some concerns	Undetected	Some concerns	No concerns	No concerns	No concerns	Moderate
01vs 03	2	Some concerns	Undetected	Some concerns	Some concerns	No concerns	No concerns	Low
01 vs 05	1	Some concerns	Undetected	Some concerns	Some concerns	No concerns	No concerns	Low
01vs 06	1	Some concerns	Undetected	Some concerns	No concerns	No concerns	No concerns	Moderate
02 vs 06	2	Some concerns	Undetected	Some concerns	Some concerns	No concerns	No concerns	Low
04 vs 06	2	Some concerns	Undetected	Some concerns	No concerns	No concerns	No concerns	Moderate
05 vs 06	4	Some concerns	Undetected	Some concerns	No concerns	No concerns	No concerns	Moderate
Success rate				Indirect evidence				
01 vs 04	-	Some concerns	Undetected	Some concerns	No concerns	No concerns	No concerns	Moderate
02 vs 03	-	Some concerns	Undetected	Some concerns	Major concerns	No concerns	No concerns	Very low
02 vs 04	-	Some concerns	Undetected	Some concerns	No concerns	No concerns	No concerns	Moderate
02 vs 05	-	Some concerns	Undetected	Some concerns	No concerns	Some concerns	No concerns	Low
03 vs 04	-	Some concerns	Undetected	Some concerns	No concerns	No concerns	No concerns	Moderate
03 vs 05	-	Some concerns	Undetected	Some concerns	Some concerns	No concerns	No concerns	Low
03 vs 06	-	Some concerns	Undetected	Some concerns	Some concerns	Some concerns	No concerns	Very low
04 vs 05	-	Some concerns	Undetected	Some concerns	No concerns	No concerns	No concerns	Moderate

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Appendix 13 Adverse events of included studies

Author Year	Sample size	Corneal edema	Hyphema	Low IOP	Shallow anterior	Vitreous hemorrhage	Choroidal detachment	Retinal detachment	Tube exposure	Tube occlusion	Encapsulated plate	Phthisis bulbi	Bleb leak	Anterior segment inflammation	Eye pain	Ocular atrophy
AGV vs AGV+IVAV																
Ma 2012	32/ 20	1/0	3/1	—	—	4/1	—	1/0	4/2	1/1	—	—	—	—	—	—
Mandy 2013	20/ 20	2/0	17/4	3/2	6/5	—	1/0	—	1/0	1/0	6/1	1/0	—	—	—	—
Zhou 2013	28/ 25	—	8/1	—	4/4	—	4/3	—	—	—	7/4	—	—	—	—	—
Acieri 2013	20/ 20	2/1	6/2	—	1/2	—	4/3	1/0	0/1	—	—	—	—	1/0	—	—
Kang 2014	13/ 14	—	—	2/1	0/1	—	1/0	—	0/2	1/2	—	—	0/2	—	—	—
Li 2019	34/ 46	4/4	8/2	2/4	1/2	—	—	—	—	8/2	—	—	—	3/4	1/2	—
AGV vs AGV+IVAV vs AGV+IVAV																

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Kong 2017	21/ 26/ 21	—	10/9/ 11	2/ 3/ 3	5/ 6/ 7	—	—	—	2/ 4/ 3	—	2/ 1/ 3	—	—	9/ 7/ 9	—	—
AGV vs CPC																
Yildiri m 2009	33/ 33	0/2	7/0	1/3	—	—	1/0	—	3/0	—	3/0	2/0	—	0/5	—	—
Liu 2014	31/ 26	10/ 12	2/8	—	—	—	—	—	—	—	—	—	—	5/6	7/ 18	0/2
Choy 2018	12/ 8	1/0	1/0	—	—	—	—	—	2/0	—	1/0	3/0	—	—	—	—
AGV+IVAV vs Trab(MMC)+IVAV																
Zhang 2014	13/ 10	—	—	—	2/1	—	—	—	—	—	—	—	—	—	—	—
Sun 2017	23/ 22	—	3/4	—	3/2	0/2	—	—	—	—	3/0	—	—	—	—	—
Gao 2018	36/ 40	—	5/3	11/9	8/4	—	4/2	—	—	—	—	—	—	—	—	—
GCT vs Trab(MMC)+IVAV																
Du	23/	15/0	7/2	—	—	—	—	—	—	—	—	—	—	9/0	16/0	—

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2015	23															
Wang 2016	30/30	16/0	6/2	—	—	—	0/1	—	—	—	—	—	0/1	7/0	20/0	—
Trab(MMC) vs Trab(MMC)+IVAV																
Saito 2010	32/20	—	20/5	—	1/0	3/2	4/4	—	—	—	—	—	1/4	—	—	—
Tanihara 2011	33/24	—	19/5	2/4	—	7/2	5/2	—	—	—	—	2/0	4/0	—	—	—
Lee 2015	26/32	—	9/5	—	—	—	8/13	—	—	—	—	—	—	—	—	—
Yu 2018	42/48	—	4/1	2/2	6/2	—	—	—	—	—	—	—	—	—	—	—
Gong 2019	82/88	3/0	7/1	1/2	3/2	—	—	—	—	—	—	—	0/1	0/1	5/2	1/0
AGV vs Trab(MMC)																
Chen 2011	20/20	—	7/8	0/2	—	—	1/1	—	—	—	—	—	0/1	—	—	—
Liu 2016	19/18	—	1/1	6/0	3/0	2/1	—	—	—	2/0	—	—	—	1/0	—	—

Notes: AGV, ahmed glaucoma valve implant surgery; CPC, cyclophotocoagulation; CCT, cyclocryotherapy; Trab, trabeculectomy; MMC, mitomycin; IVAV, intravitreal anti- vascular endothelial growth factor; IOP, intraocular pressure reduction.

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Appendix 14 Composition of the primary diseases in the included studies

References	Study design	Interventions	DR	RVO	OIS	CRAO	Others	Combine PRP
Ma,2012	OCT	AGV vs AGV+IVAV	26 vs 15	5 vs 5	1 vs 0	—	—	Yes
Mahdy,2013	RCT	AGV vs AGV+IVAV	15 vs 16	4 vs 4	1 vs 0	—	—	Yes
Zhou,2013	OCT	AGV vs AGV+IVAV	12 vs 7	10 vs 13	1 vs 2	2 vs 1	3 vs 2	Yes
Kang,2014	OCT	AGV vs AGV+IVAV	9 vs 12	2 vs 1	—	—	2 vs 1	Yes
Arcieri,2015	RCT	AGV vs AGV+IVAV	11 vs 10	9 vs 10	—	—	—	Yes
Kwon,2017	OCT	AGV vs AGV+IVAV	14 vs 33	3 vs 5	5 vs 7	—	—	Yes
Li,2019	OCT	AGV vs AGV+IVAV	14 vs 20	20 vs 26	—	—	—	N/A
Kong,2017	OCT	AGV vs AGV+IVAV vs AGV+IVAV	10 vs 12 vs12	10 vs 11 vs 7	1 vs 3 vs 2	—	—	N/A
Yildirim,2009	RCT	AGV vs CPC	NA	NA	NA	NA	NA	NA
Liu,2014	OCT	AGV vs CPC	13 vs 12	11 vs 9	—	—	7 vs 5	NA
Choy,2018	RCT	AGV vs CPC	6 vs 4	4 vs 1	1 vs 3	1 vs 0	—	Yes
Zhang,2014	OCT	AGV+IVAV vs Trab(MMC)+IVAV	5 vs 3	5 vs 5	—	—	3 vs 2	Yes
Sun,2017	OCT	AGV+IVAV vs Trab(MMC)+IVAV	—	—	—	—	—	Yes
Gao,2018	OCT	AGV+IVAV vs Trab(MMC)+IVAV	16 vs 16	16 vs 22	3 vs 1	1 vs 1	—	Yes

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Du,2015	OCT	CCT vs Trab(MMC)+IVAV	19	17	—	—	10	Yes
Wang,2016	OCT	CCT vs Trab(MMC)+IVAV	14 vs 14	14 vs 13	—	—	4 vs 3	Yes
Saito,2010	OCT	Trab(MMC) vs Trab(MMC)+IVAV	22 vs 14	6 vs 6	4 vs 0	—	—	Yes
Takihara,2011	OCT	Trab(MMC) vs Trab(MMC)+IVAV	28 vs 18	4 vs 6	1 vs 0	—	—	Yes
Lee,2015	OCT	Trab(MMC) vs Trab(MMC)+IVAV	24 vs 25	2 v 4	0 vs 3	—	—	Yes
Yu,2018	OCT	Trab(MMC) vs Trab(MMC)+IVAV	16 vs 18	26 vs 30	—	—	—	Yes
Song,2019	OCT	Trab(MMC) vs Trab(MMC)+IVAV	43 vs 43	39 vs 45	—	—	—	NA
Shen,2011	OCT	AGV vs Trab(MMC)	14 vs 13	4 vs 3	1 vs 1	2 vs 2	—	NA
Liu,2016	OCT	AGV vs Trab(MMC)+IVAV	8 vs 6	9 vs 10	1 vs 0	—	—	NA

Notes: AGV, ahmed glaucoma valve implant surgery; CPC, cyclophotocoagulation; CCT, cyclocryotherapy; Trab, trabeculectomy; MMC, mitomycin; IVAV, intravitreal anti- vascular endothelial growth factor; DR, diabetic retinopathy; RVO, retinal vein occlusion; OIS, ocular ischemic syndrome; CRAO, central retinal artery occlusion; PRP, panretinal photocoagulation.

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MOOSE Checklist for Meta-analyses of Observational Studies

Item No	Recommendation	Reported on Page No
Reporting of background should include		
1	Problem definition	Page 6-8
2	Hypothesis statement	Page 8
3	Description of study outcome(s)	Page 8
4	Type of exposure or intervention used	Page 7-8
5	Type of study designs used	Page 7-8
6	Study population	Page 6-8
Reporting of search strategy should include		
7	Qualifications of searchers (eg, librarians and investigators)	Page 27
8	Search strategy, including time period included in the synthesis and key words	Page 8 & Appendix
9	Effort to include all available studies, including contact with authors	Page 8
10	Databases and registries searched	Page 8
11	Search software used, name and version, including special features used (eg, explosion)	Page 8
12	Use of hand searching (eg, reference lists of obtained articles)	None
13	List of citations located and those excluded, including justification	Page 8
14	Method of addressing articles published in languages other than English	Page 8
15	Method of handling abstracts and unpublished studies	Page 9
16	Description of any contact with authors	Page 9
Reporting of methods should include		
17	Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested	Page 9
18	Rationale for the selection and coding of data (eg, sound clinical principles or convenience)	Page 9-10
19	Documentation of how data were classified and coded (eg, multiple raters, blinding and interrater reliability)	Page 9-10
20	Assessment of confounding (eg, comparability of cases and controls in studies where appropriate)	Page 9-10
21	Assessment of study quality, including blinding of quality assessors, stratification or regression on possible predictors of study results	Page 10
22	Assessment of heterogeneity	Page 11
23	Description of statistical methods (eg, complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated	Page 11
24	Provision of appropriate tables and graphics	Page 11
Reporting of results should include		
25	Graphic summarizing individual study estimates and overall estimate	Page 11-20,31 & Appendix
26	Table giving descriptive information for each study included	Page 12 & Table 2
27	Results of sensitivity testing (eg, subgroup analysis)	Page 18 & Appendix
28	Indication of statistical uncertainty of findings	Page 18 & Appendix

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Item No	Recommendation	Reported on Page No
Reporting of discussion should include		
29	Quantitative assessment of bias (eg, publication bias)	Page 21-23,25
30	Justification for exclusion (eg, exclusion of non-English language citations)	Page 21-22
31	Assessment of quality of included studies	Page 23,25
Reporting of conclusions should include		
32	Consideration of alternative explanations for observed results	Page 25
33	Generalization of the conclusions (ie, appropriate for the data presented and within the domain of the literature review)	Page 25
34	Guidelines for future research	Page 25
35	Disclosure of funding source	Page 4,27

From: Stroup DF, Berlin JA, Morton SC, et al, for the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) Group. Meta-analysis of Observational Studies in Epidemiology. A Proposal for Reporting. *JAMA*. 2000;283(15):2008-2012. doi: 10.1001/jama.283.15.2008.

Transcribed from the original paper within the NEUROSURGERY® Editorial Office, Atlanta, GA, United Sates. August 2012.

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page 3-4
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 6-7
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 7-8
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 9
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 8
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Page 8 & Appendix
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 8-9
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 10
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 9-10
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 9-10
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page 10
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Page 10-11
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Page 11
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Page 11
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Page 11
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Page 11
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Page 11
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Page 11
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Page 11
Certainty	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Page 11

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 11 & Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Page 11
Study characteristics	17	Cite each included study and present its characteristics.	Page 12 & Table 2
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Page 14-15 & Appendix
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Page 15-21,31
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Page 15-17,19-20,31 & Table 5 & Appendix
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Page 15-17,19-20,31 & Table 5 & Appendix
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Page 18 & Appendix
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Page 18 & Appendix
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Page 18 & Appendix
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Page 18 & Appendix
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page 20-24
	23b	Discuss any limitations of the evidence included in the review.	Page 25
	23c	Discuss any limitations of the review processes used.	Page 25
	23d	Discuss implications of the results for practice, policy, and future research.	Page 26
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page 4
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Page 4
	24c	Describe and explain any amendments to information provided at registration or in the protocol. (Trial registration and protocol: None.)	
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 4, 27

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Competing interests	26	Declare any competing interests of review authors.	Page 2, 27
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Page 26 & Appendix

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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PRISMA 2020 for Abstracts Checklist

Section and Topic	Item #	Checklist item	Reported (Yes/No)
TITLE			
Title	1	Identify the report as a systematic review.	Yes
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Yes
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Yes
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Yes
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	Yes
Synthesis of results	6	Specify the methods used to present and synthesise results.	Yes
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Yes
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Yes
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	Yes
Interpretation	10	Provide a general interpretation of the results and important implications.	Yes
OTHER			
Funding	11	Specify the primary source of funding for the review.	Yes
Registration	12	Provide the register name and registration number.	No

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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